FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2022/2023

Subject : Bioinformatics I (SECB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 02 – Access to Biological Databases

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1. The purpose of this problem is to introduce you to using Entrez and related NCBI resources.

a)	How many human proteins are bigger than 300,000 daltons?
	5214 human proteins are bigger than 300,000 daltons.
b)	What is the longest human protein?
	The longest human protein is Titin.

There are several different ways to solve these questions.

(1) Try to first limit your search to human by using TaxBrowser. From the home page of NCBI select the alphabetical list of resources and find the Taxonomy Browser and the entry for human. Then follow the link to Entrez Protein, where all the results will be limited to human.

(2) Enter a command in the format xxxxxx:yyyyyy[molwt] to restrict the output to a certain number of Daltons; for example, 002000:010000[molwt] will select proteins of molecular weight 2,000 to 10,000.

(3) As a different approach, search 30000:50000[Sequence Length]

(4) You can read more about titin (NP_596869.4), the longest human protein, in an NCBI newsletter (WebLink 2.73 ftp://ftp.ncbi.nih.gov/genbank/gbrel.txt). While the average protein has a length of several hundred amino acids, incredibly titin is 34,423 amino acids in length.

(5) Explore additional ways to limit Entrez searches by using an NCBI Handbook chapter (http://www.ncbi.nlm.nih.gov/books/NBK44864/)

Write a report of how you would answer the question in this week's session.

Include screen captures as well as detailed explanation and observations.

All screen captures should be labelled as figures and numbered accordingly.

All tables should be labelled as tables and numbered accordingly.

Hint: What is the Latin name for humans? Homo sapiens.

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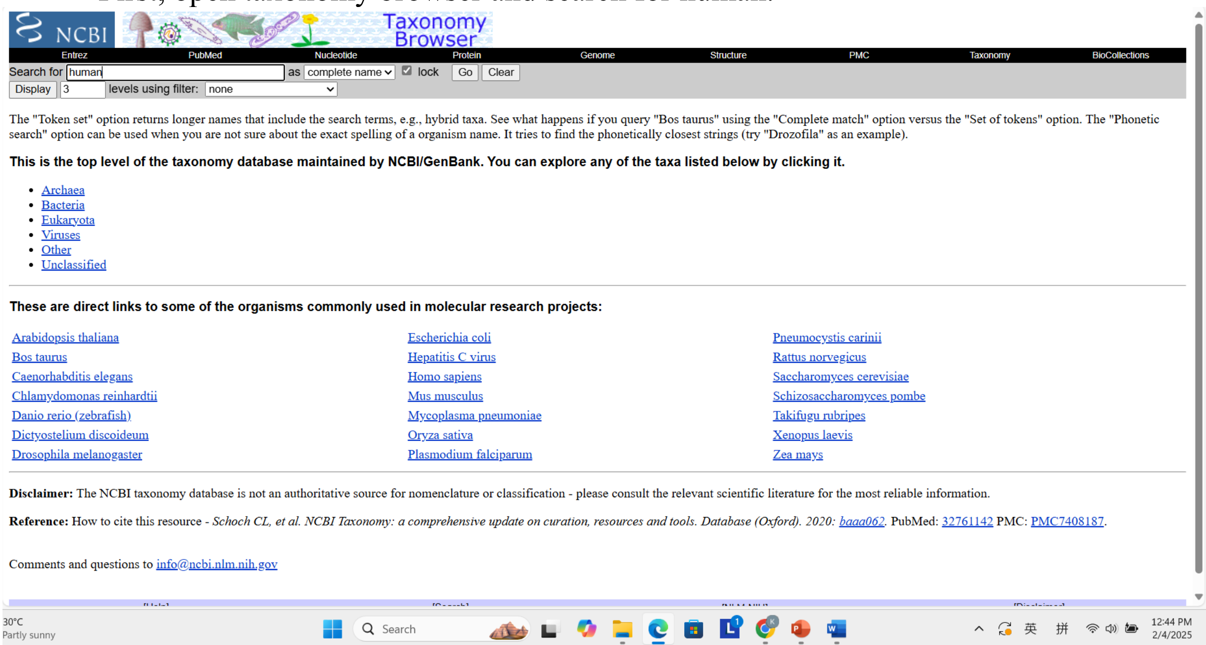
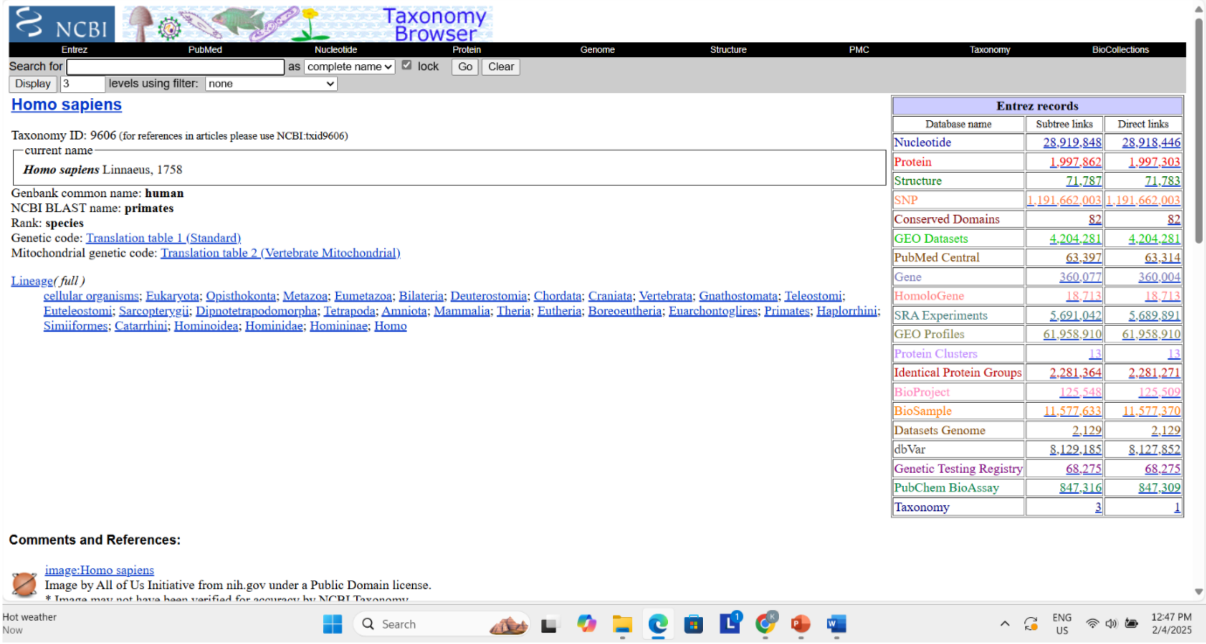
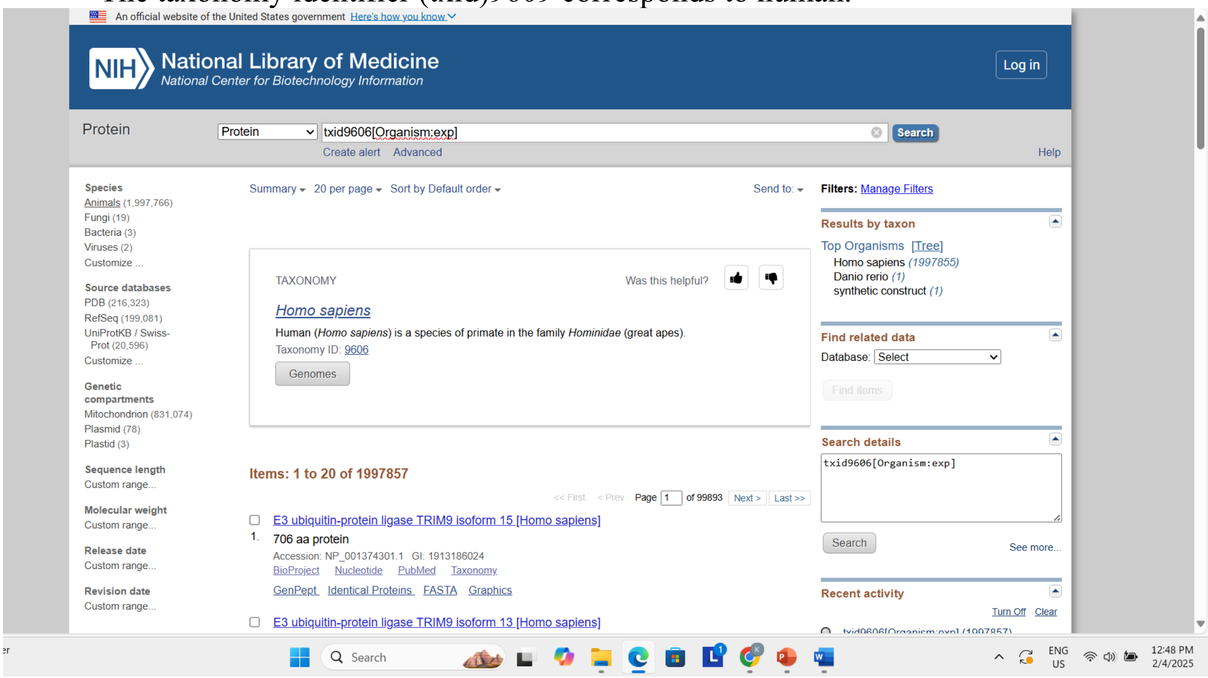
1. Answer :
   First, open taxonomy browser and search for human.



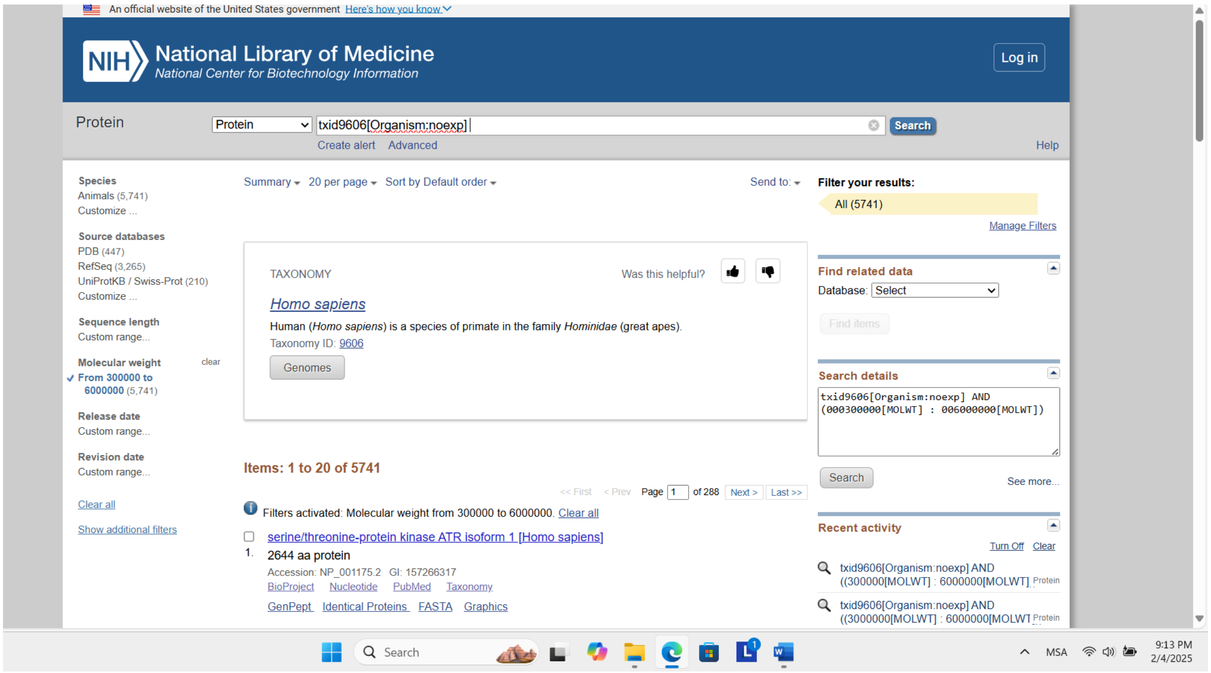
Then, click on homo sapiens.



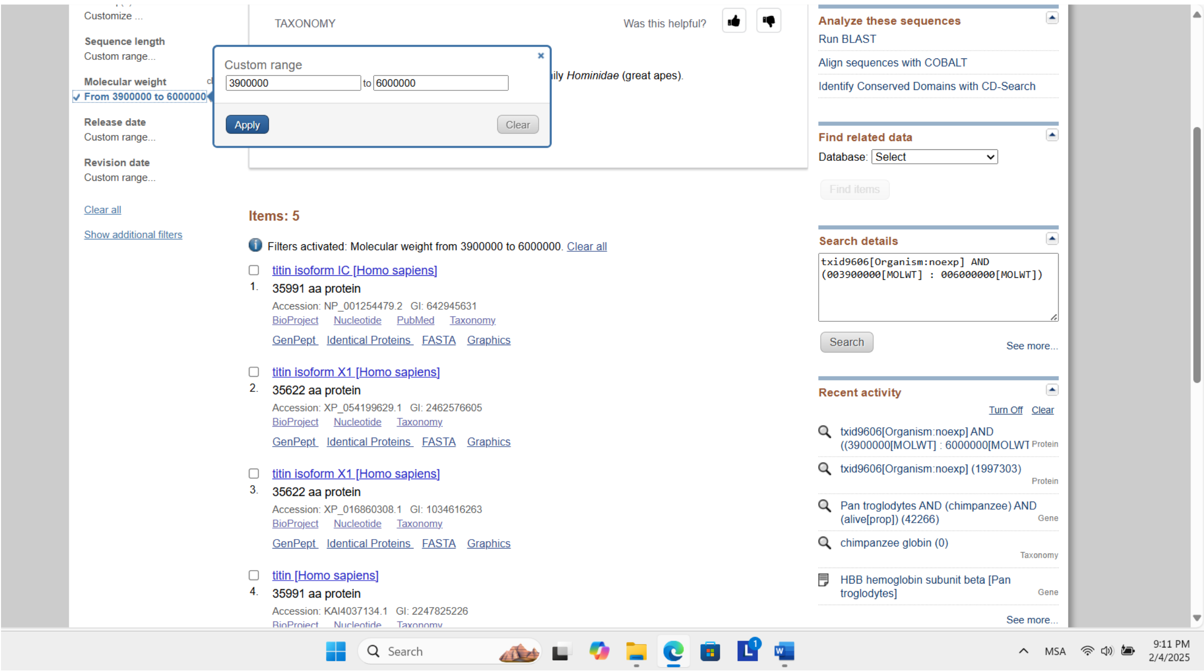
Next, on the above page, click the protein link. This allows to link to proteins that are human. The taxonomy identifier (txid)9609 corresponds to human.



On the left sidebar select the molecular weight range from 300,000 to 600,000.



This show the result, there are 5741 proteins > 3000,000 daltons.



Here shows titan is the longest human protein.

2. The purpose of this problem is to obtain information from the NCBI website.

The RefSeq accession number of human beta globin protein is NP_000509. Go to NCBI

(http://www.ncbi.nlm.nih.gov/).

a) What is the RefSeq accession number of beta globin protein from the chimpanzee (Pan troglodytes)?

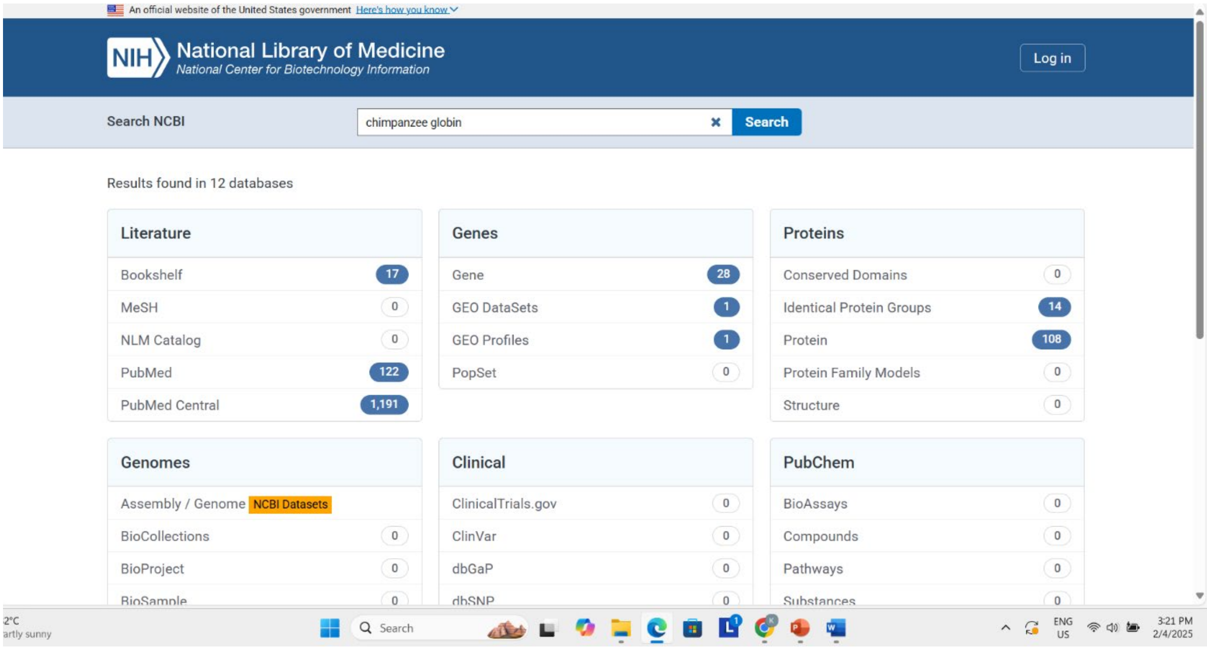
XP_508242.1

(1) There are several different ways to solve this. Try typing chimpanzee globin into the home page of NCBI; or use the Taxonomy Browser to find chimpanzee Entrez Gene entries.

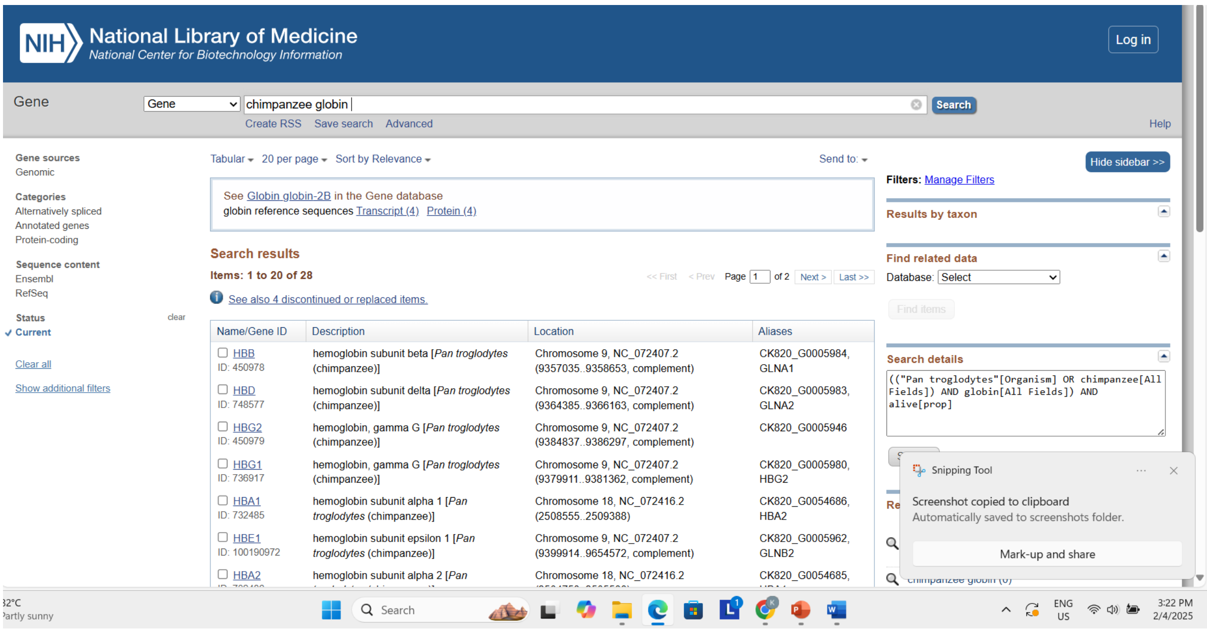
(2) HomoloGene (http://www.ncbi.nlm.nih.gov/homologene)(WebLink 2.38) is a great resource to learn about sets of related eukaryotic proteins. Use HomoloGene to find a set of beta globins including chimpanzee.

2. Answer

Entering chimpanzee globin into the NCBI home page leads to results in Gene



The Gene result sorts chimpanzee HBB to the top:



Follow the first entry (HBB) to see the RefSeq accession. In the above screen shot you can also see the accession NC_072407.2. This corresponds to the entire chromosome 9 of the chimp.

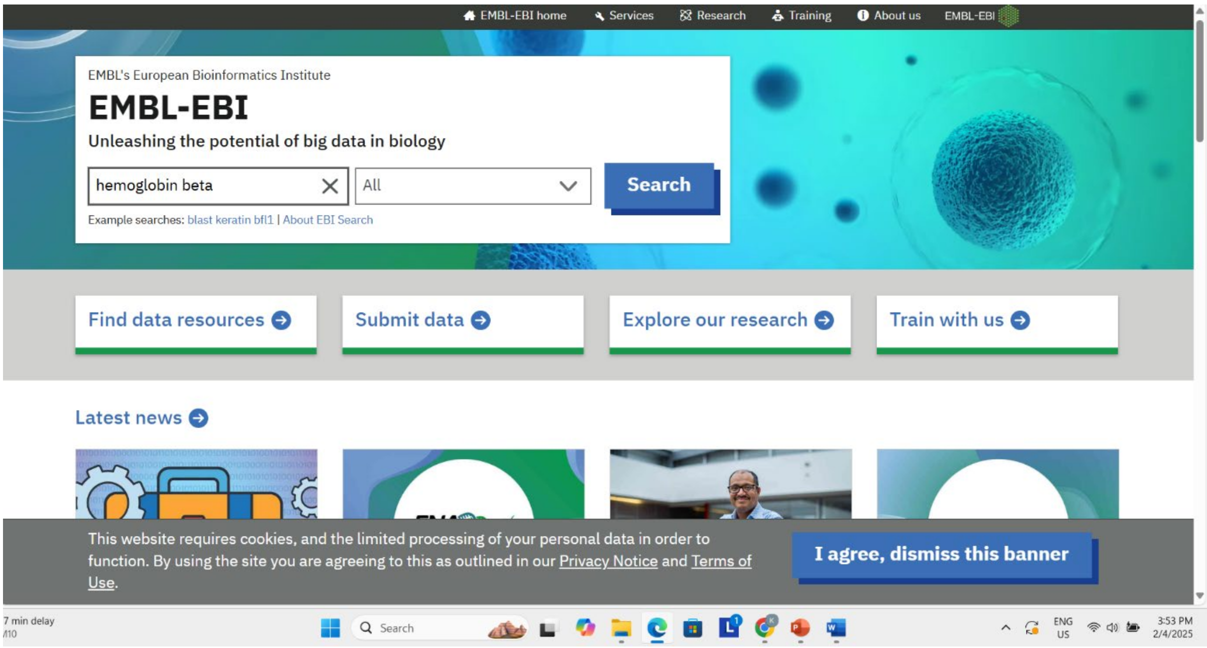
3. The purpose of this exercise is to become familiar with the EBI website and how to use it to access information.

(1) Visit the site (http://www.ebi.ac.uk/)(WebLink 2.5). Enter hemoglobin beta in the main query box (alternatively use the query human hemoglobin beta).

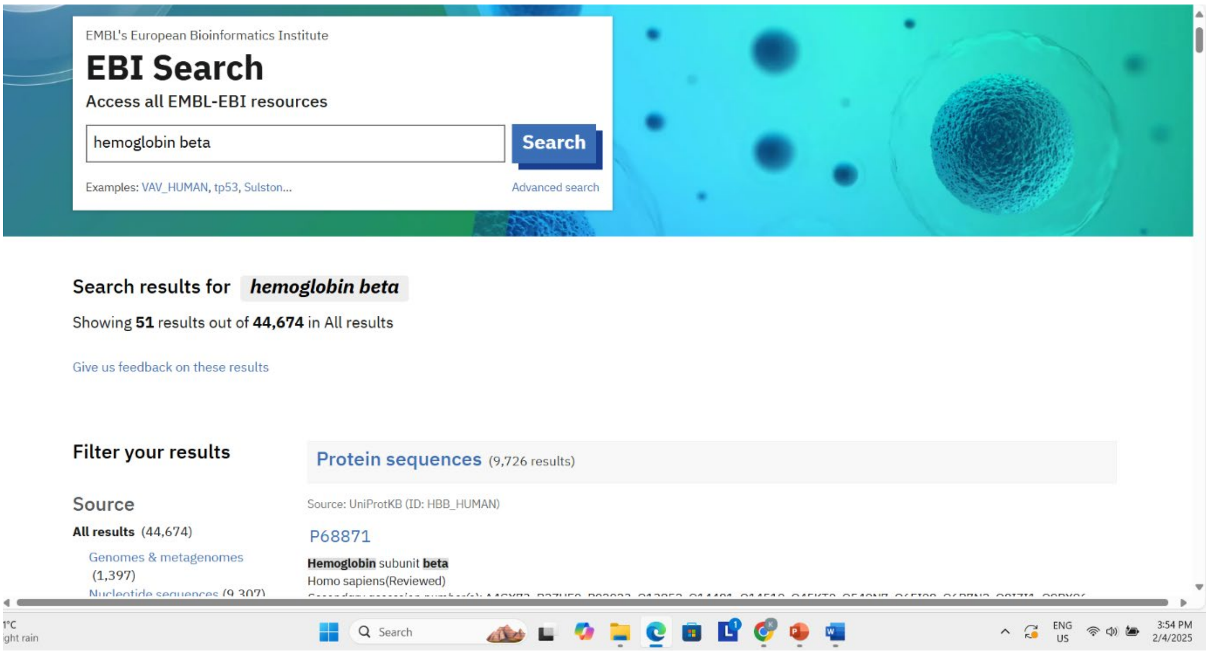
(2) Inspect the results. Explore the various links to information about pathways, genomes, nucleotide and protein sequences, structures, protein families, and more.

3.Answer

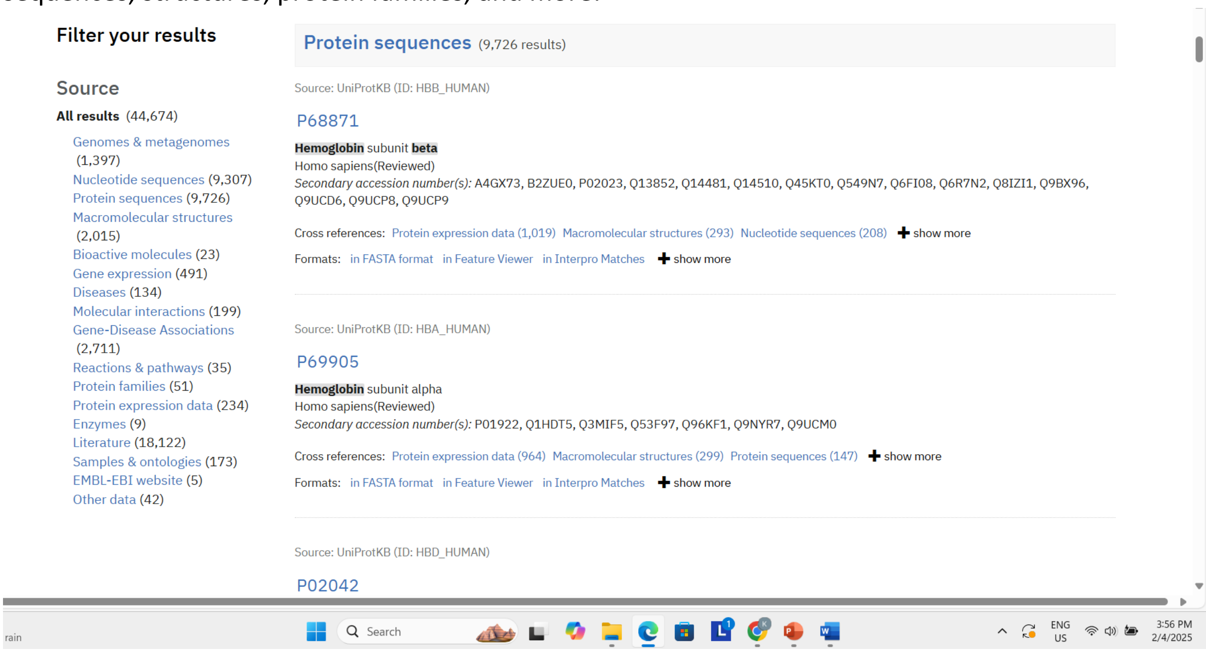
Enter hemoglobin beta at the EBI home page:



Here is the search result page:



Here can explore the various links to information about pathways, genomes, nucleotide and protein sequences, structures, protein families, and more:



4. Accessing information from BioMart: the beta globin locus

(1) Go to www.ensembl.org and follow the link to BioMart.

(2) First choose a database; we will select Ensembl Genes 71.

(3) Choose a dataset: Homo sapiens genes (GRCh37.p10). Note the other available datasets.

(4) Choose a filter. Here the options include region, gene, transcript event, expression, multispecies comparisons, protein domains, and variation. Select “region”, chromosome 11, and enter 5240000 for the Gene Start (bp) and 5300000 (bp) for the Gene End. (Note that this region spans 60 kilobases and corresponds to chr11:5,240,001-5,300,000.)

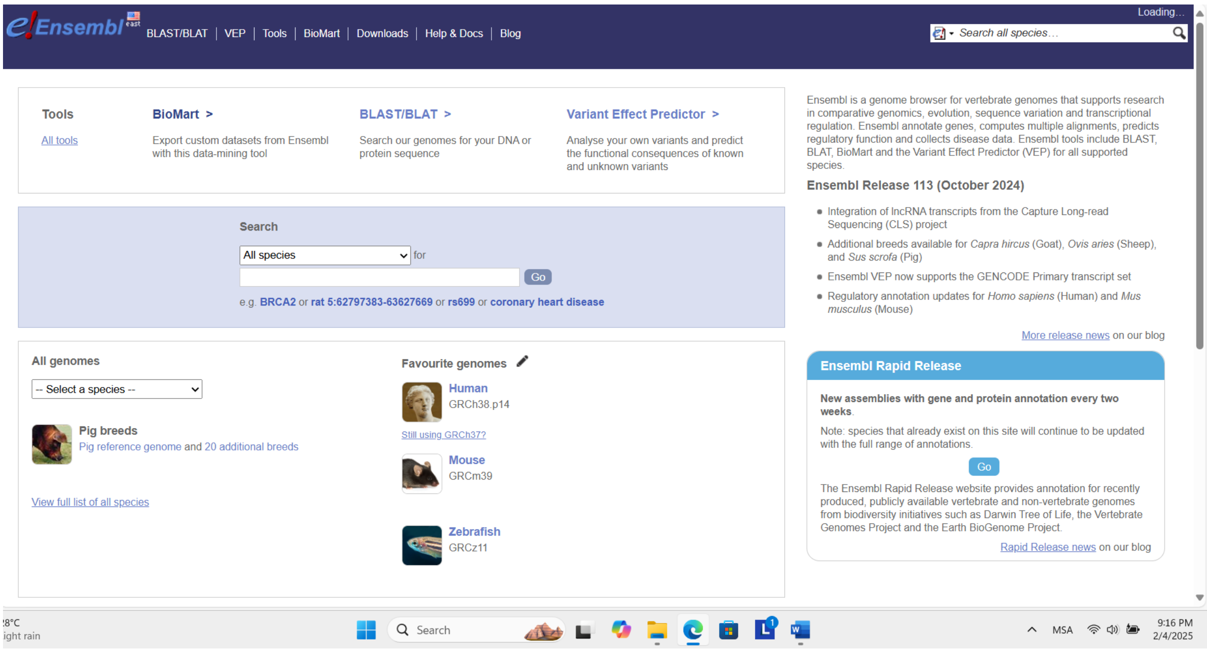
(5) Choose attributes. Select the following features. Under “Gene” select Ensembl Gene ID and %GC content; under “External” select the external references CCDS ID, HGNC symbol (this is the official gene symbol) and HGNC ID(s).

(6) At the top left select “Count.” Currently there are 8 genes matching these criteria.

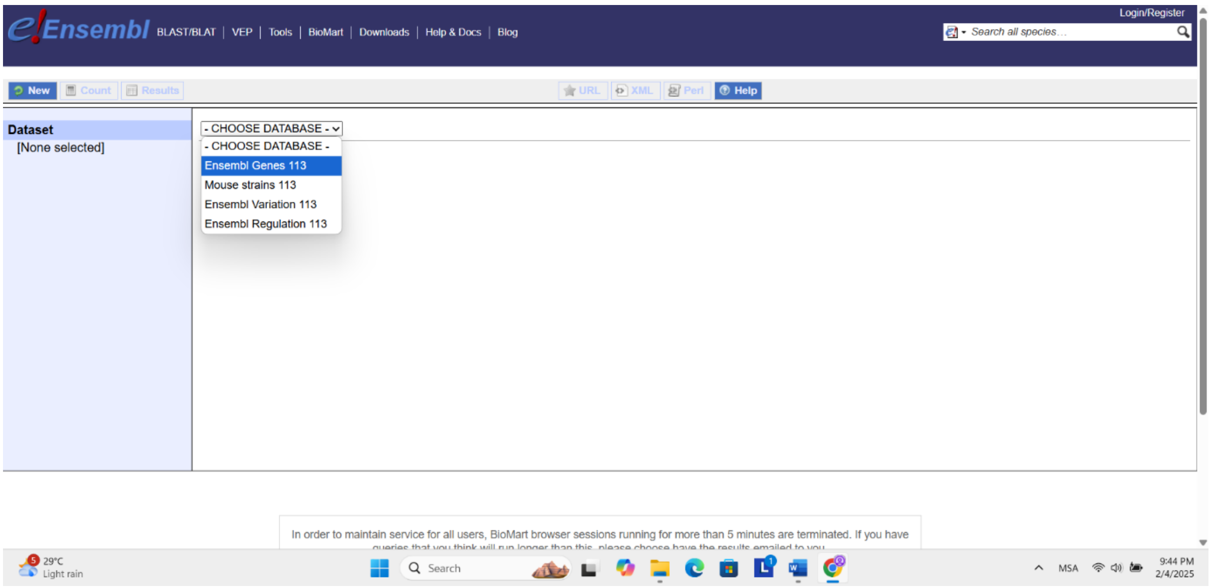
(7) To view these results press “Results. “Note that you can export your results in several formats (including a comma separated values or CSV file) that can be further manipulated (e.g., converted to a BED file).

4.Answer

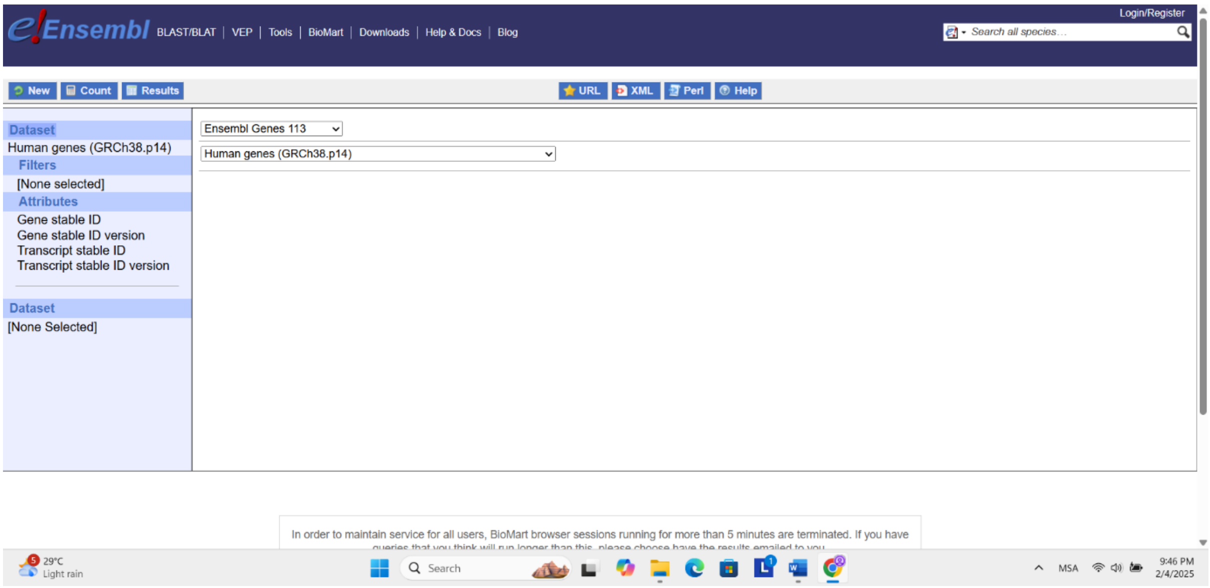
This is the homepage of the link:



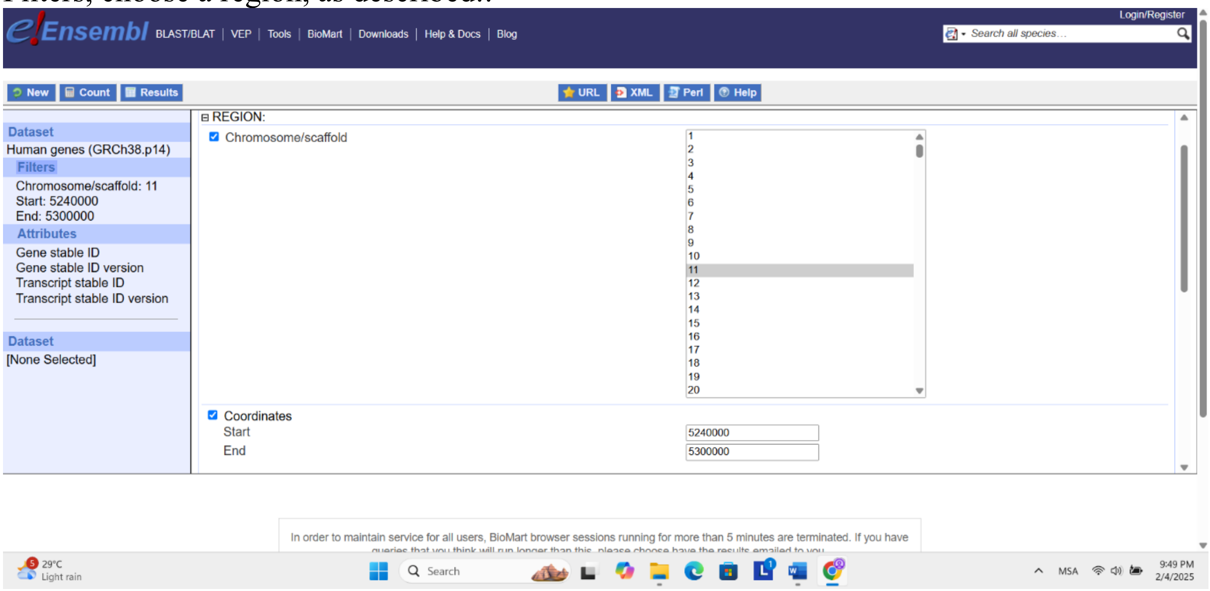
Choose a database (Ensembl Genes 113)



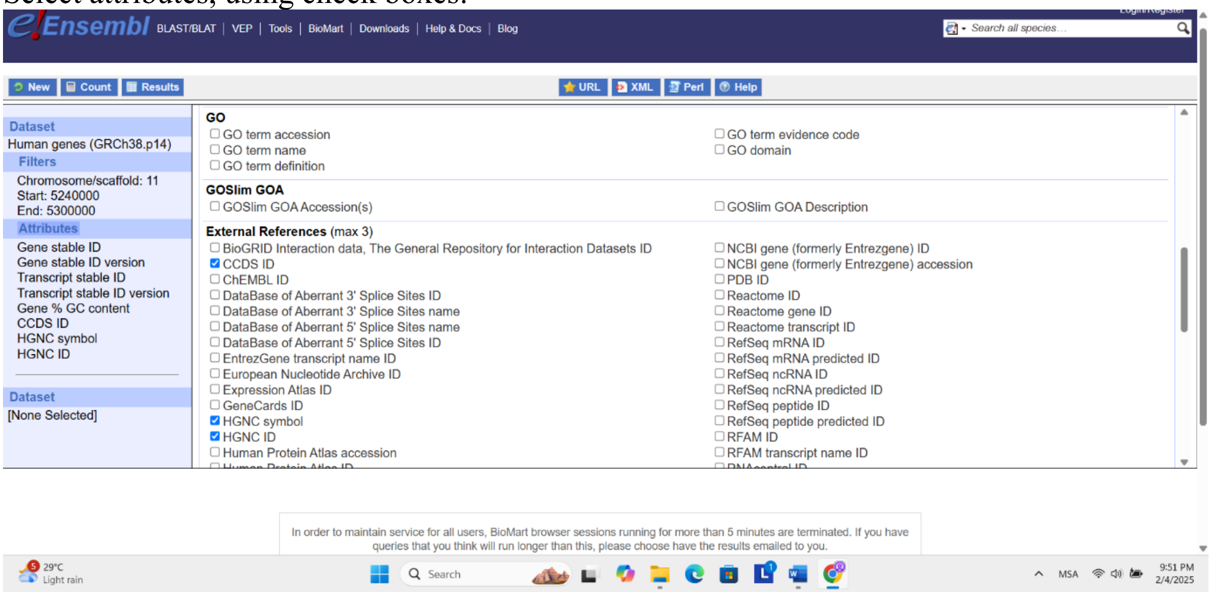
Choose dataset:



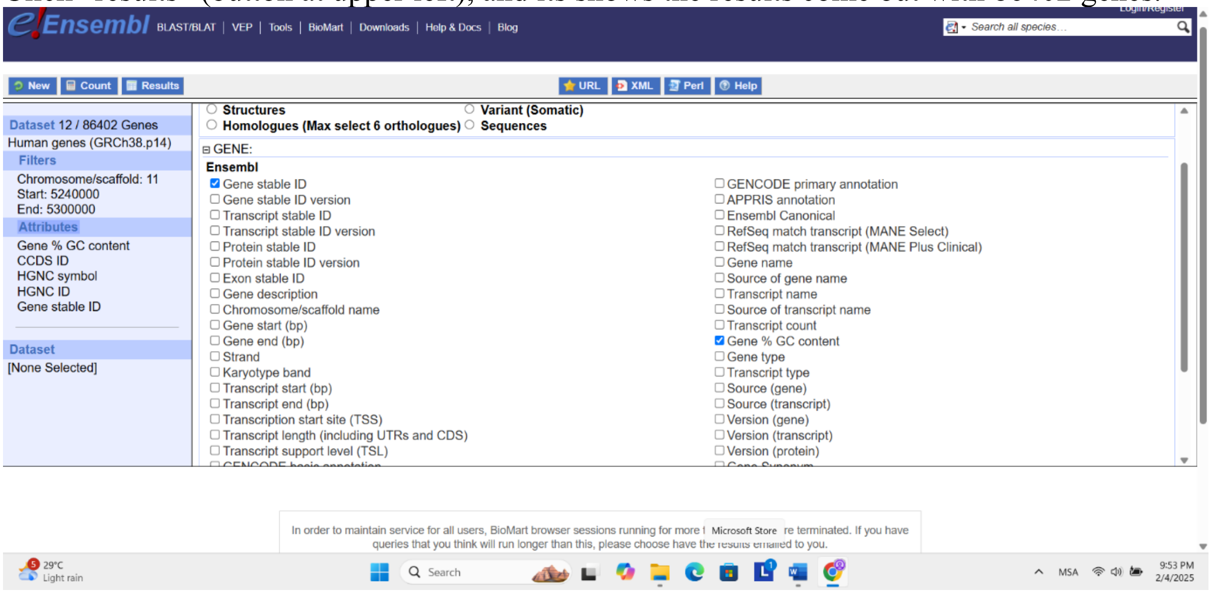
Filters, choose a region, as described::



Select attributes, using check boxes:



Click “results” (button at upper left), and its shows the results come out with 86402 genes.



5. **BioMart: working with lists. The goal of this exercise is to access information in BioMart by uploading a text file listing gene identifier of interest.**

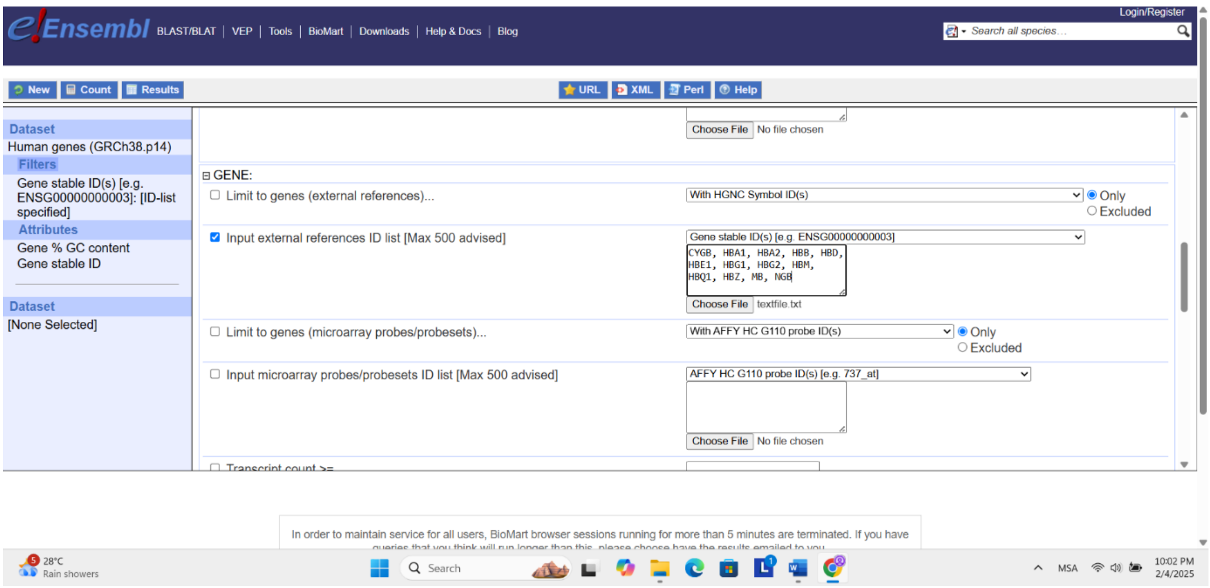
Follow the steps from problem 2-4, but for the filter set choose Gene (instead of Region), select ID list limit and adjust the pulldown menu to HGNC symbol, then browse for a text file having a list of gene symbols.

See Web Document 2.5 (WebDocument_2-5_13_humanGlobins_HGNCsymbols.txt) for a text file listing official HGNC symbols for 13 human globin genes (*CYGB*, *HBA1*, *HBA2*, *HBB*, *HBD*, *HBE1*, *HBG1*, *HBG2*, *HBM*, *HBQ1*, *HBZ*, *MB*, *NGB*).

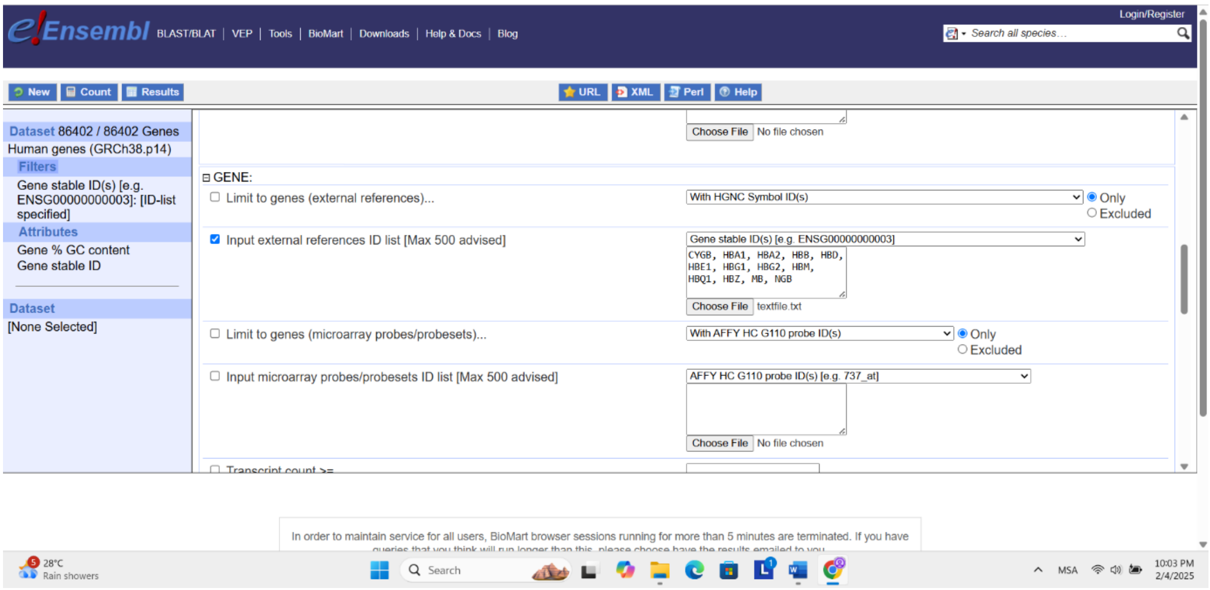
You could also enter these gene symbols manually.

For attributes choose any set of features that is different than in problem 2-4, so that you can further explore BioMart resources.

5.Answer



Results: (86402genes match)


Ensembl
BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog
Login/Register
Search all species...
New
Count
Results
URL
XML
Perl
Help
Dataset 86402 / 86402 Genes
Human genes (GRCh38.p14)
Filters
Gene stable ID(s) [e.g. ENSG00000000003]: [ID-list specified]
Attributes
Gene % GC content
Gene stable ID
Dataset
[None Selected]
Choose File No file chosen
GENE:
Limit to genes (external references)...
With HGNC Symbol ID(s)
Only
Excluded
Input external references ID list [Max 500 advised]
Gene stable ID(s) [e.g. ENSG00000000003]
CYGB, HBA1, HBA2, HBB, HBD, HBE1, HBG1, HBG2, HBM, HBQ1, HBZ, MB, NGB
Choose File textfile.txt
Limit to genes (microarray probes/probesets)...
With AFFY HC G110 probe ID(s)
Only
Excluded
Input microarray probes/probesets ID list [Max 500 advised]
AFFY HC G110 probe ID(s) [e.g. 737_at]
Choose File No file chosen
Transcript count >=
In order to maintain service for all users, BioMart browser sessions running for more than 5 minutes are terminated. If you have
28°C
Rain showers
Search
MSA
10:03 PM
2/4/2025

6. Accessing information from Ensembl.

(1) Visit the Ensembl resource for humans (via http://www.ensembl.org/human).

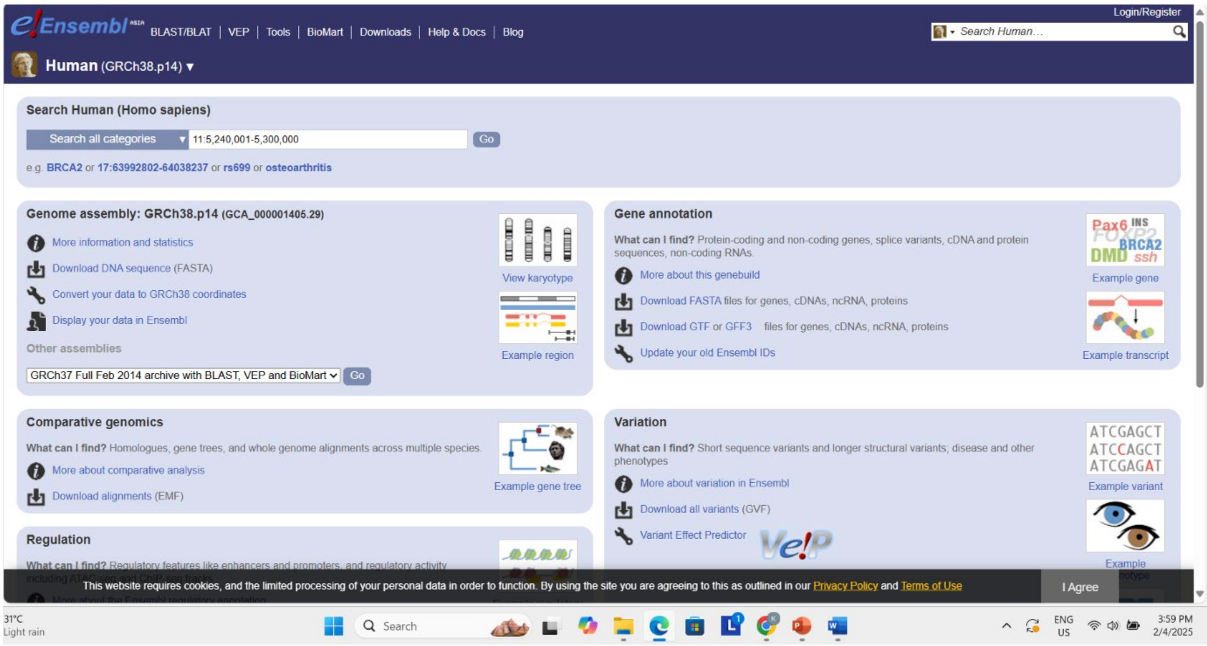
(2) In the main search box enter 11:5,240,001-5,300,000. The resulting page displays several panels. At the top, all of chromosome 11 is shown. Where on the chromosome is the region we have selected? In what chromosomal band does this region reside?

(3) The next panel shows the region in detail. What is the size of the displayed region, in base pairs? In general, genes encoding olfactory receptors are gamed OR followed by a string of numbers and letters (e.g., *OR51F1*). Approximately how many olfactory receptor genes flank the 60 kb region we have selected? Can you determine exactly how many ORs are in that region?

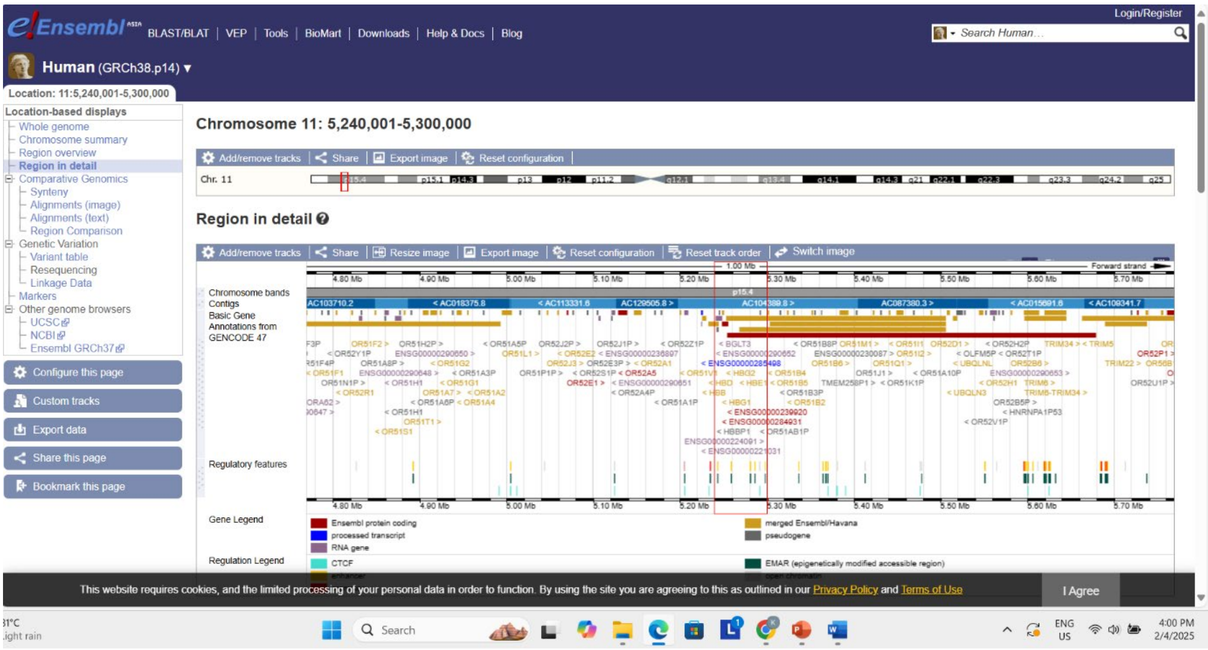
(4) Next, we see the region we selected (11:5240001-5300000). Note that there are horizontal tracks (similar to the UCSC Genome Browser).

6.Answer

At the search box, enter 11:5,240,001-5,300,000 at human categories.



Here is the upper portion of the results page:



Here shows many olfactory receptor (OR) genes.

7. Accessing information from UCSC

Hemoglobin is a tetramer composed primarily of two alpha globin subunits and two beta globin subunits.

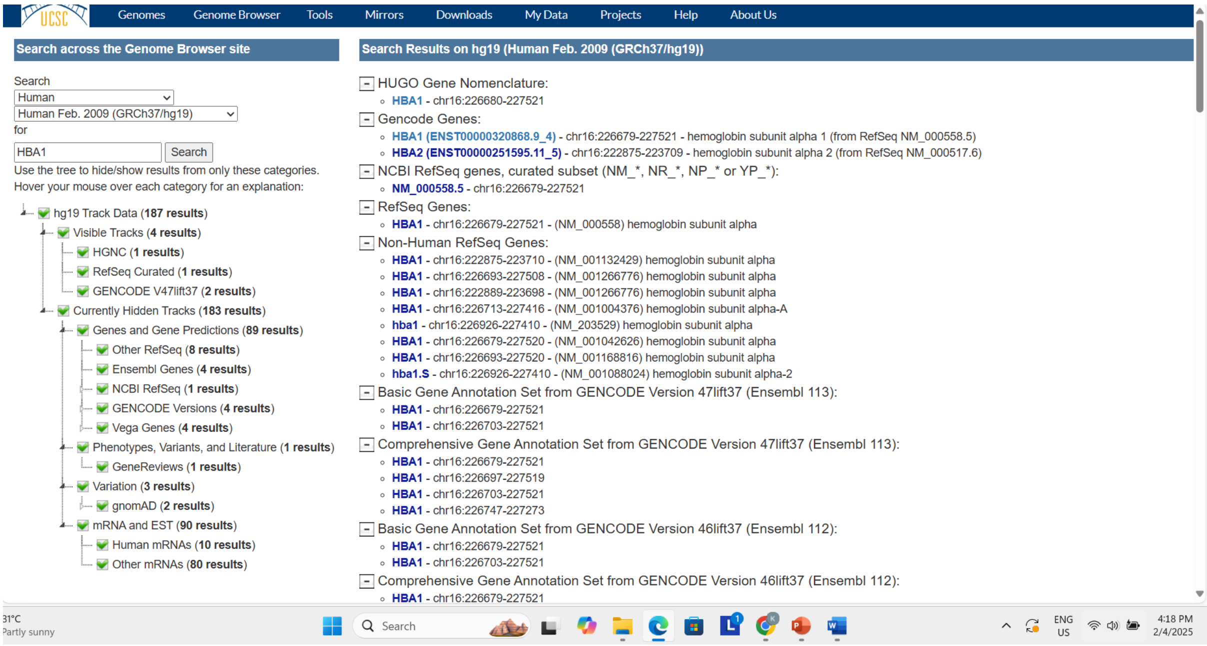
Consider alpha globin.

There are two related human genes (official gene symbols HBA1 and HBA2).

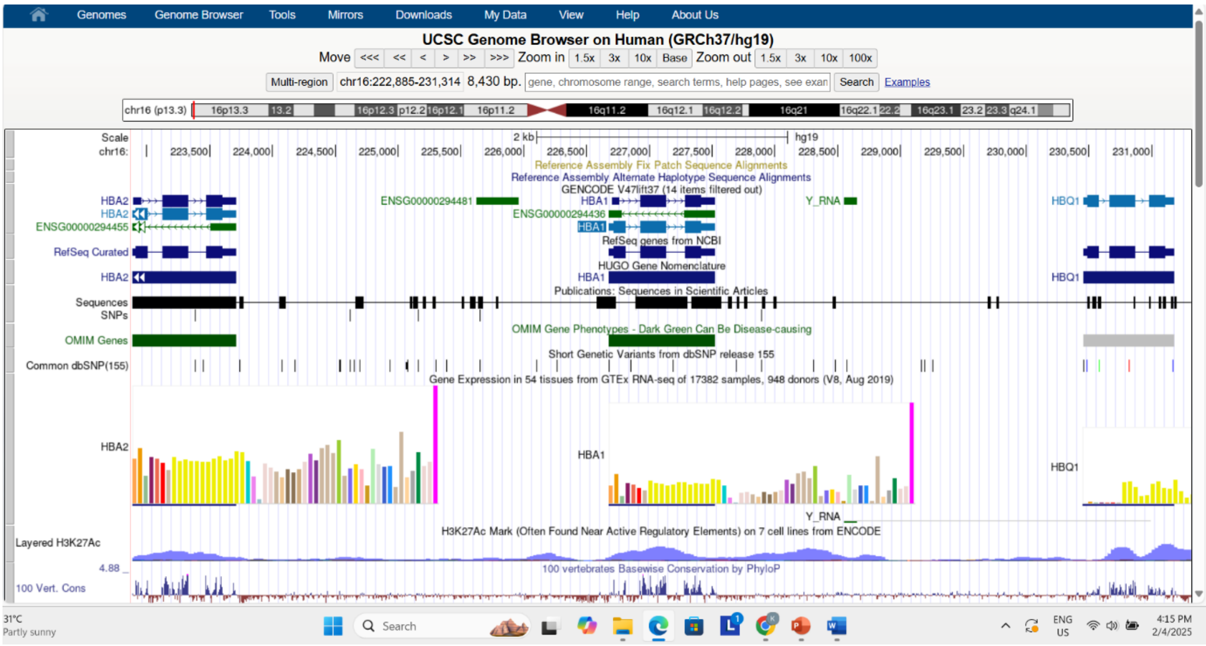
a) Use the UCSC Genome Browser (http://genome.ucsc.edu/) to determine the length of the intergenic region between the HBA1 and HBA2 genes.

7.Answer
Enter HBAI at the search bar:

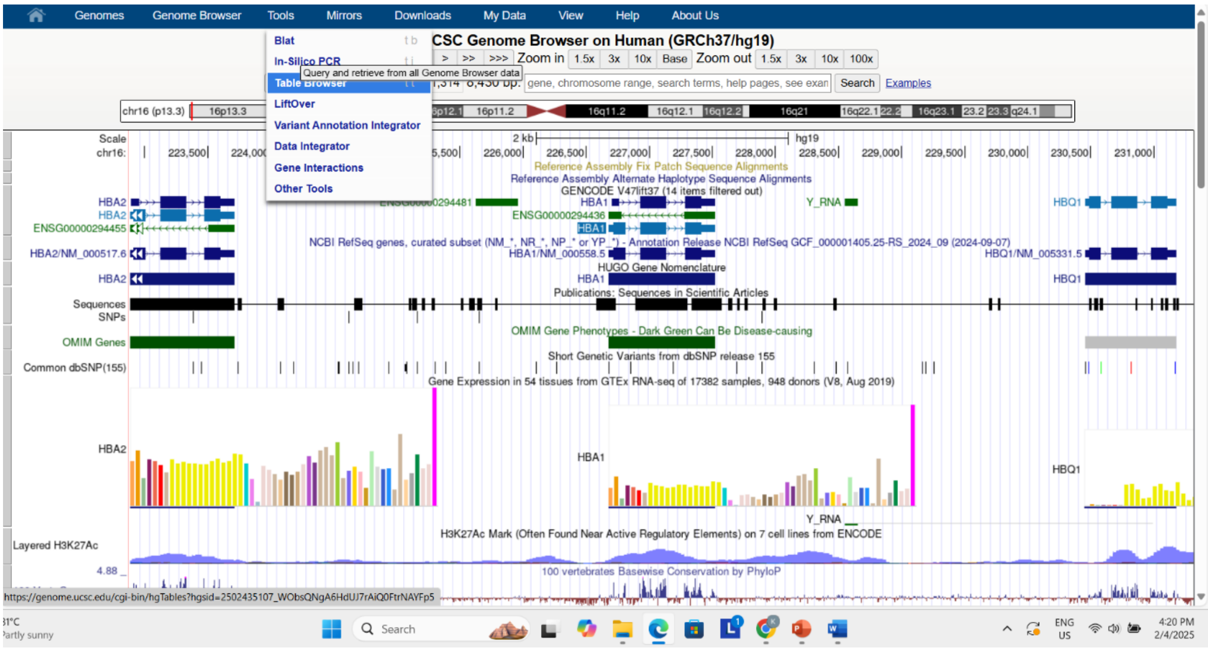
It will show:



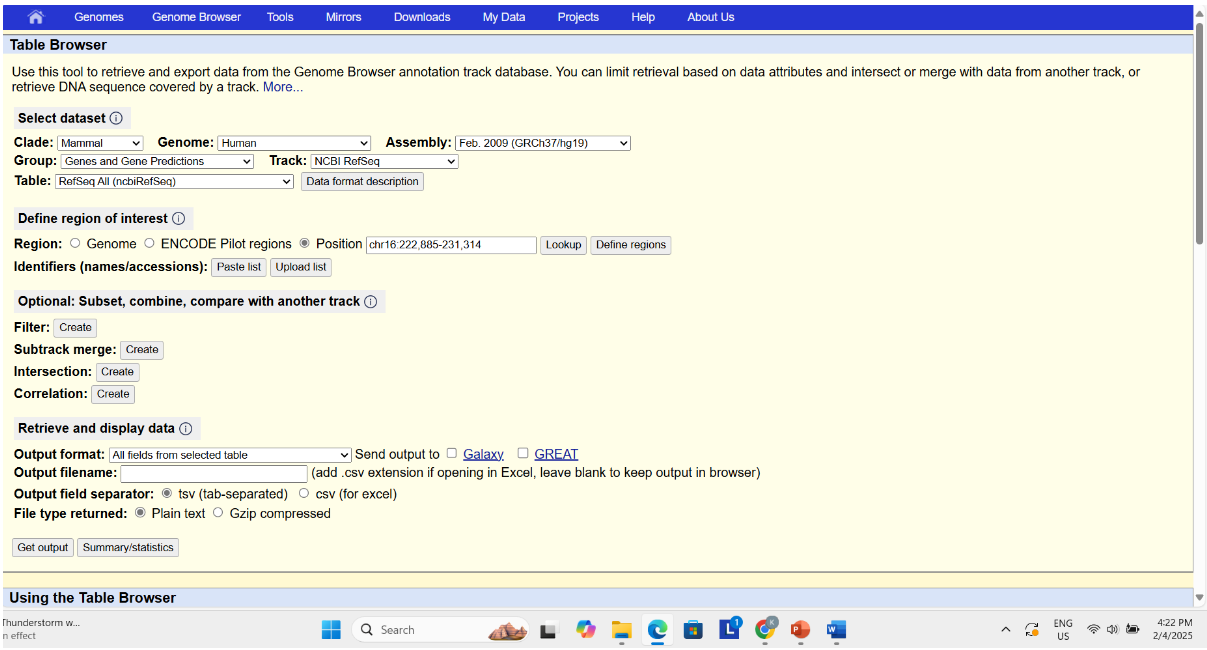
Click on HBA1 and zoom out 10-fold and you will see the adjacent HBA1 and HBA2 genes:



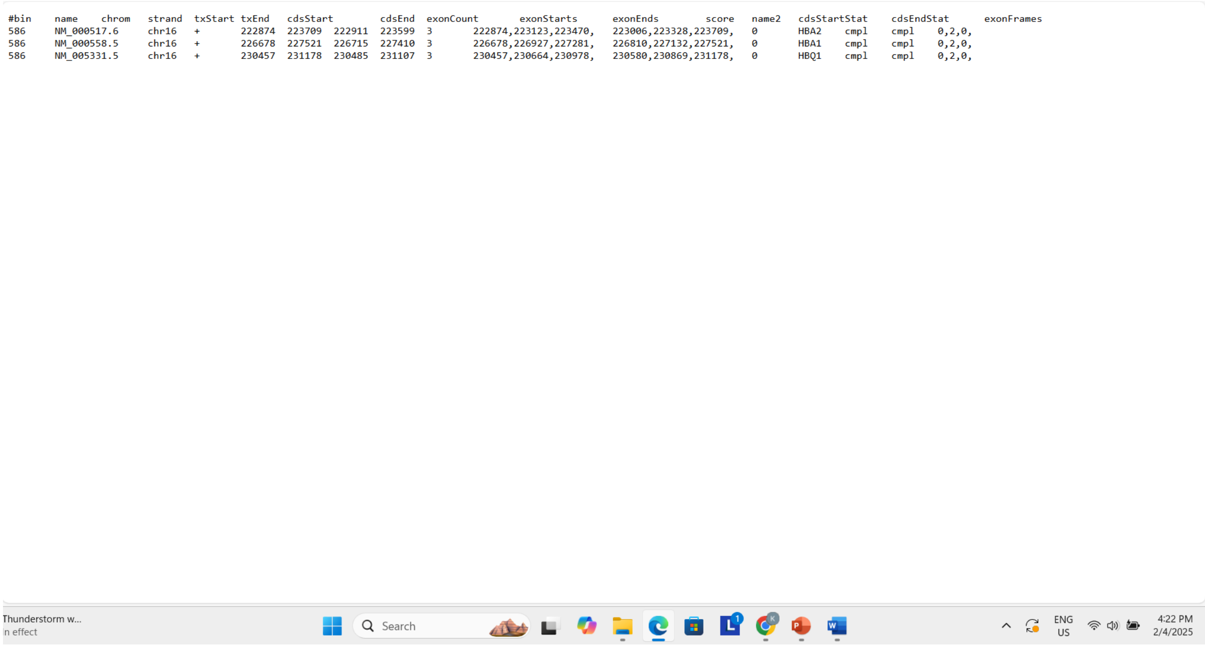
To get a more precise answer use the Table Browser, from the Tool menu:



Set a region encompassing both genes and set the various menus as shown here:

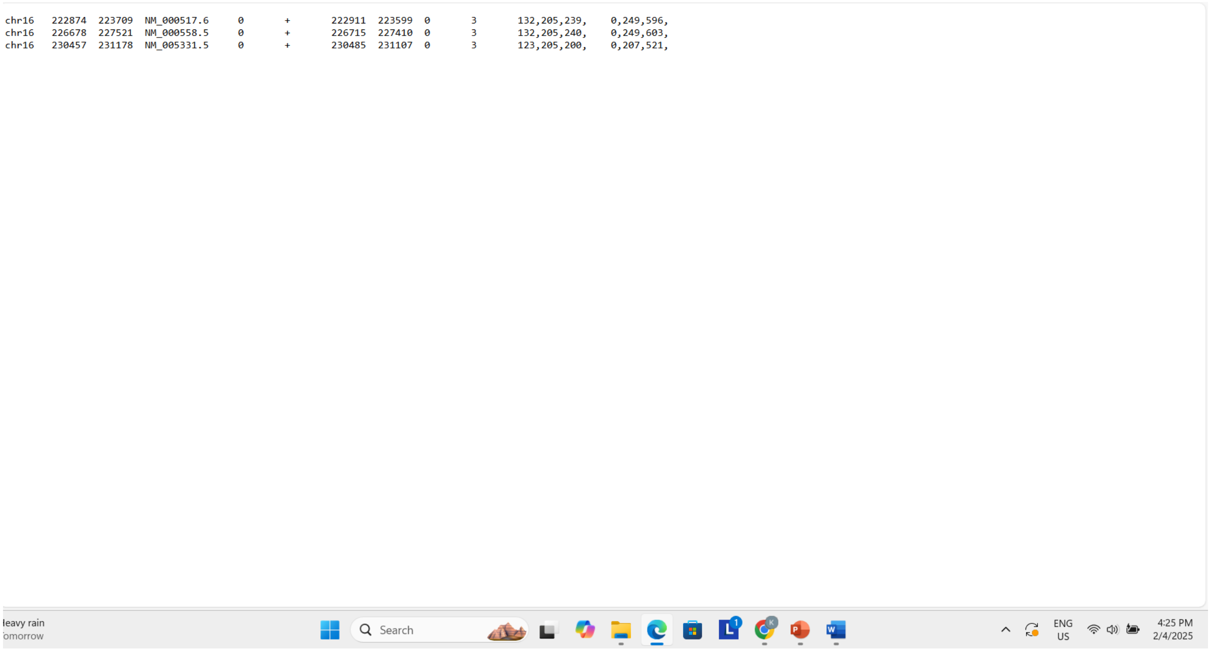


This is the output (set to all fields):



#bin	name	chrom	strand	txStart	txEnd	cdsStart	cdsEnd	exonCount	exonStarts	exonEnds	score	name2	cdsStartStat	cdsEndStat	exonFrames
586	NM_000517.6	chr16	+	222874	223709	222911	223599	3	222874,223123,223470,	223006,223328,223709,	0	HBA2	cmpl	cmpl	0,2,0,
586	NM_000558.5	chr16	+	226678	227521	226715	227410	3	226678,226927,227281,	226810,227132,227521,	0	HBA1	cmpl	cmpl	0,2,0,
586	NM_005331.5	chr16	+	230457	231178	230485	231107	3	230457,230664,230978,	230580,230869,231178,	0	HBQ1	cmpl	cmpl	0,2,0,

This is the output (as a BED file). From either of these outputs you can calculate the exact intergenic distance.



chr16	222874	223709	NM_000517.6	0	+	222911	223599	0	3	132,205,239,	0,249,596,
chr16	226678	227521	NM_000558.5	0	+	226715	227410	0	3	132,205,240,	0,249,603,
chr16	230457	231178	NM_005331.5	0	+	230485	231107	0	3	123,205,200,	0,207,521,

Take the beginning of one the more 3’ gene (position 226,715) and subtract the end of the more 5’ gene (position 223,599) = 3,116 base pairs.

8. Accessing information from UCSC. The purpose of this exercise is for you to gain familiarity with the UCSC Genome Browser

a) What types of repetitive DNA elements occur in the human beta globin gene?

The purpose of this exercise is for you to gain familiarity with the UCSC Genome Browser.

As a user, you choose which tracks to display. Visit and explore as many as possible. Try to get a sense for the main categories of information offered at the Genome Browser.

As you work in the genome browser you may want to switch between builds hg18, hg19, and hg20. To do so, go the "View" pull-down menu and use "In other genomes (convert)."

Follow the following steps.

(1) Go to http://genome.ucsc.edu/cgi-bin/hgGateway . Make sure the clade is Mammal, genome is Human, assembly is NCBI36/hg19, and in the "gene" box enter hbb for beta globin. Click Submit. Note that HBB is the official gene symbol for beta globin, but you can use the lowercase hbb for this search. Use NCBI Gene (or http://www.genenames.org for the HGNC site) to find the official gene symbol of your favorite gene.

(2) Click the "default tracks" button. Note the position you have reached (chromosome 11, spanning 1,606 base pairs close to the beginning of the short or "p" arm of the chromosome). Note the appearance of over a dozen graphical tracks that are horizontally oriented.

(3) One of the tracks is “Repeating Elements by Repeatmasker.” There are two black blocks. Right click on the block and select “Full.” Alternatively, scroll down to the section entitled “Variation and Repeats,” locate “RepeatMasker,” and change the pull-down menu setting from “dense” to “full.” Note also that by clicking the blue heading “RepeatMasker” you visit a page describing the RepeatMasker program and its use at the UCSC Genome Browser.

(4) View the RepeatMasker output.

Choose one answer.

a. There are no repetitive elements.

b. There is one SINE element and one LINE element.

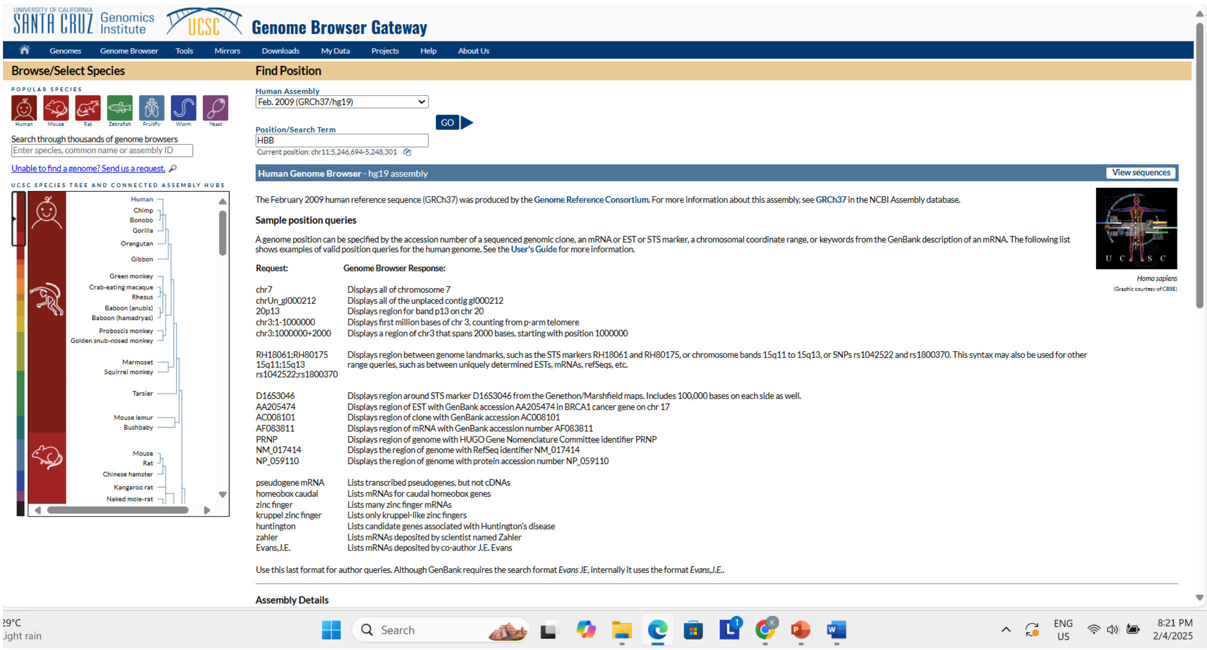
c. There is one LTR and one satellite.

d. There is one LINE element and one low complexity element.

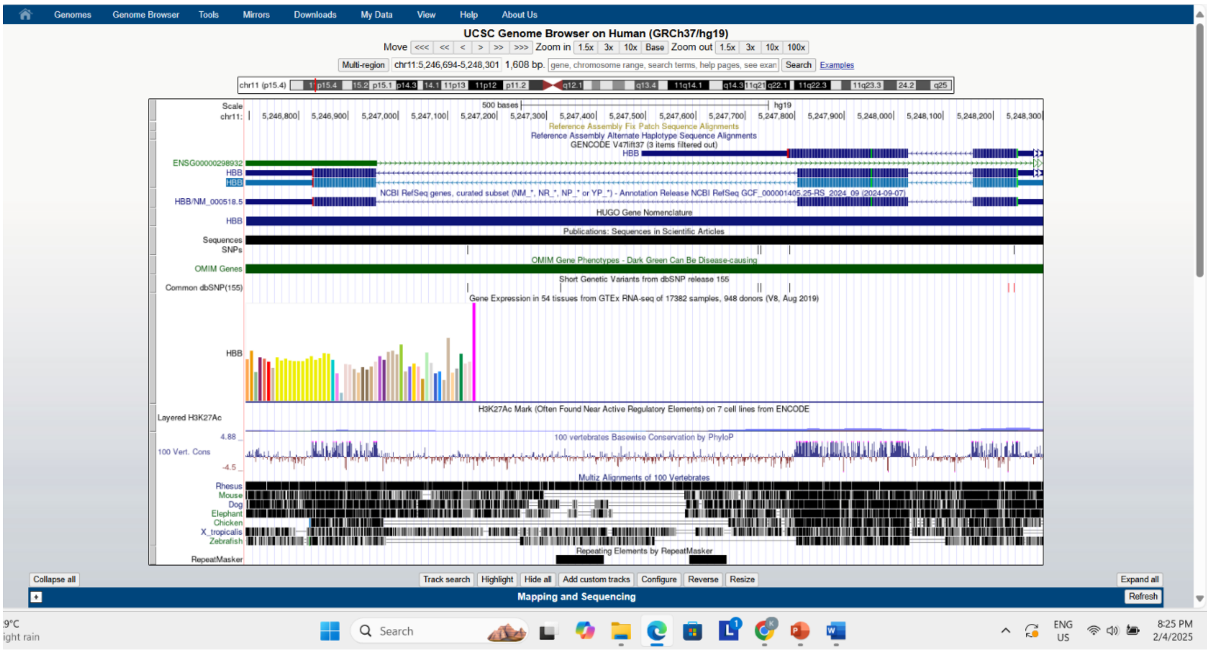
e. There are well over a dozen repetitive elements.

8.Answer:

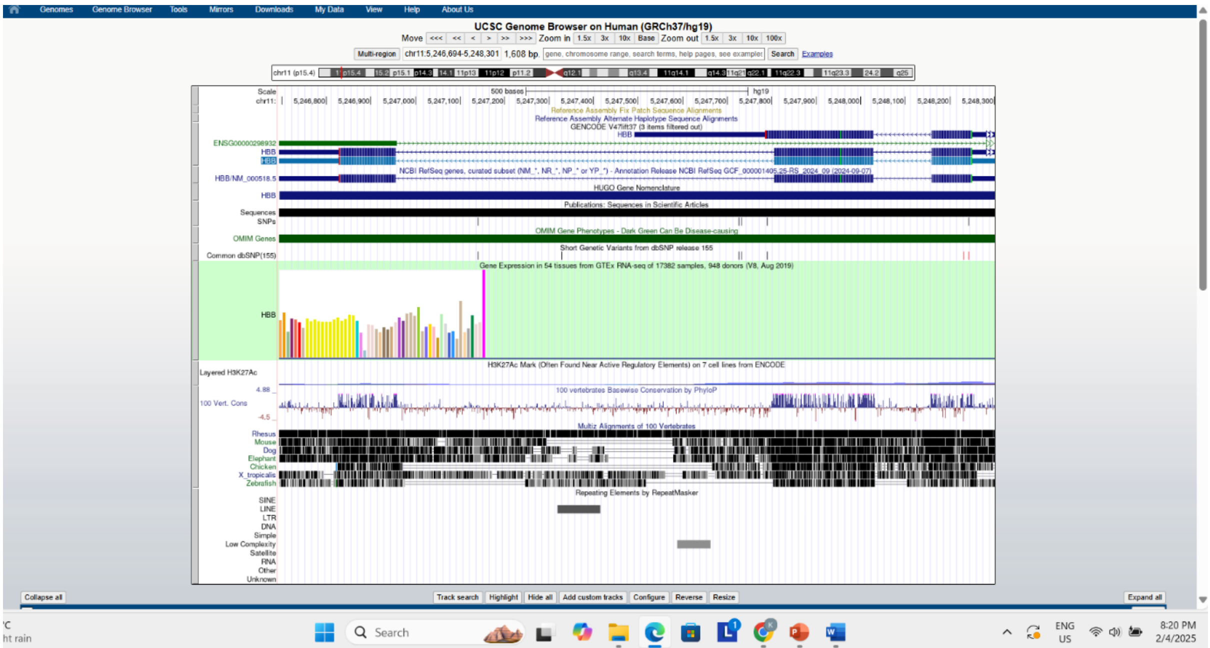
Enter HBB in search bar:



The view of default tracks:



The view with the RepeatMasker track(at bottom) expanded:



By inspection answer (d) is correct.

9. Accessing information from the UCSC Table Browser.

a) How many SNPs span the human beta globin gene?

To solve this problem, use the UCSC Table Browser. The Table Browser is equally useful as the Genome Browser. Instead of offering visual output, it offers tabular output. Often it is not practical (or accurate) to visually count elements from the Genome Browser. We often want quantitative information about genomic features in some chromosomal region or across the whole genome. This problem asks about single nucleotide polymorphisms (SNPs), which are positions that vary (i.e., exhibit polymorphism) across individuals in a population. Follow the following steps.

(1) Start at the HBB region of the UCSC Genome Browser and click the "Tables" tab along the top. Alternatively, you can go to the UCSC website (http://genome.ucsc.edu), and click Tables. Set the clade to Mammal, genome to Human, assembly to NCBI36/hg18, group to Variation and Repeats, track to SNPs(130), table to snp130, and region to position chr11:5203272-5204877. Note that if the position is not already set, you can type hbb into the position box, click "lookup" and the correct position will be entered.

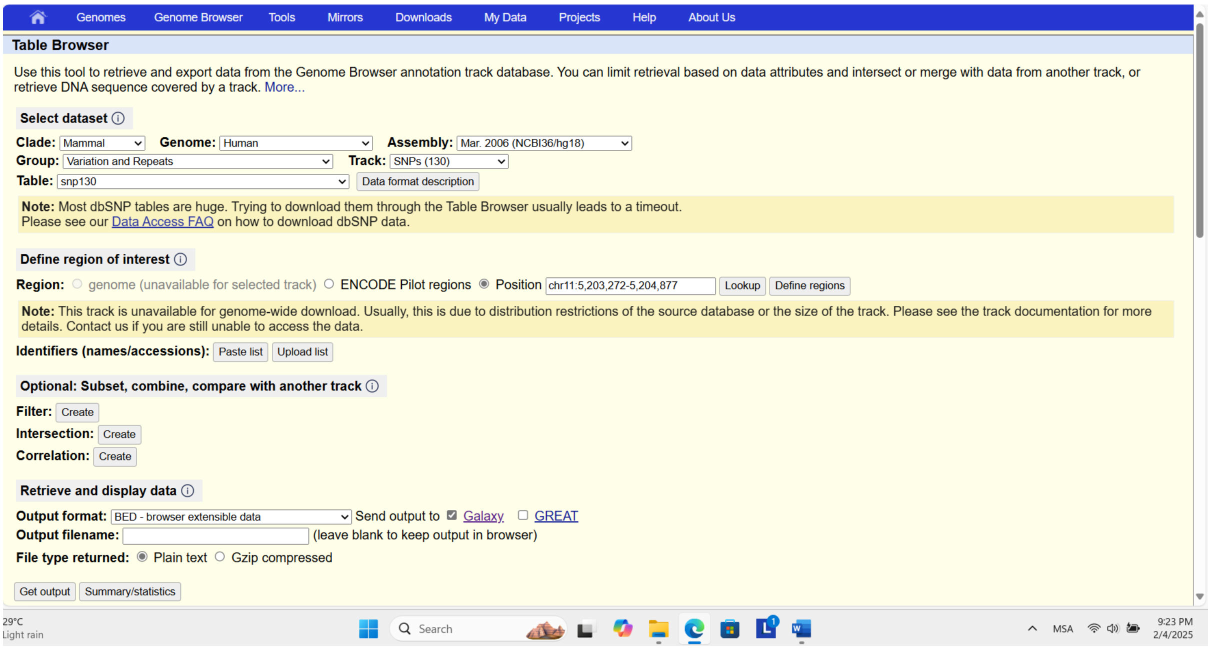
(2) To see the answer to this problem, click "summary/statistics." The item count tells you how many SNPs there are.

(3) To see the answer as a table, set the output format to "all fields from selected table," make sure the "Send output to Galaxy / GREAT" boxes are not checked, and click the "get output" box. The SNPs are shown as a table including chromosome, start, and stop position.

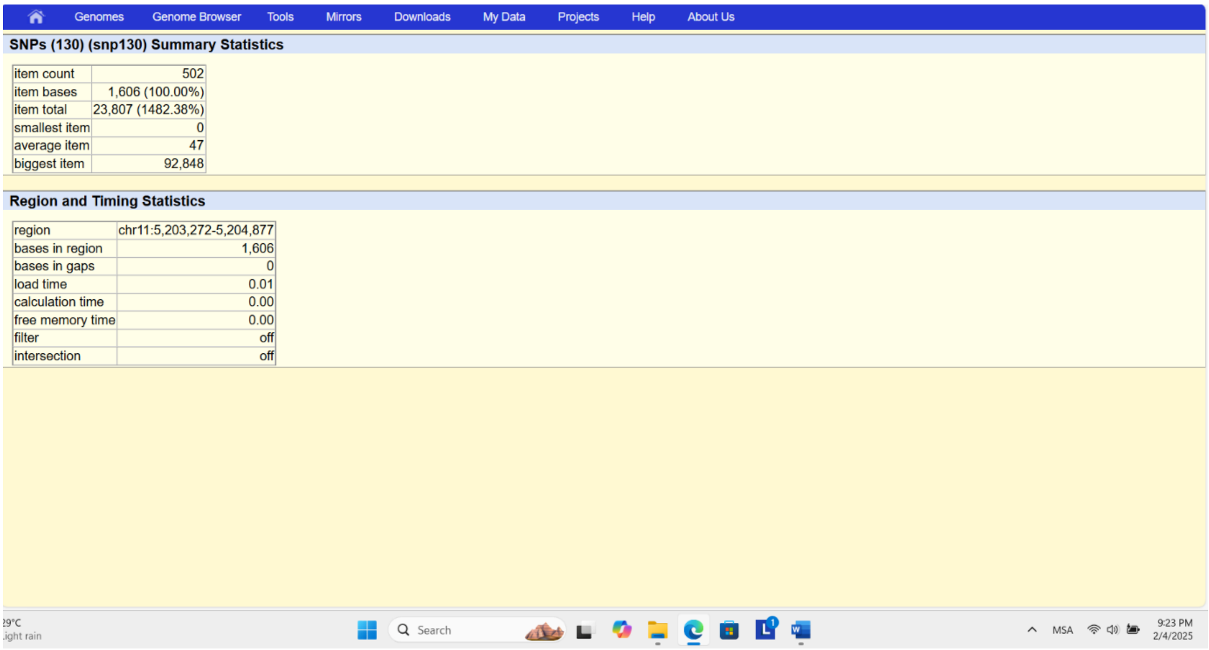
(4) Try the various output options, such as a bed file or a custom track. Note that you can output the information as a file saved to your computer.

9.Answer:

In the group "Variation and repeat" the tracks that are currently available include SNPs(130):



For SNPs(130), clicking summary/statistics shows :



item count	502
item bases	1,606 (100.00%)
item total	23,807 (1482.38%)
smallest item	0
average item	47
biggest item	92,848

region	chr11:5,203,272-5,204,877
bases in region	1,606
bases in gaps	0
load time	0.01
calculation time	0.00
free memory time	0.00
filter	off
intersection	off

10. Accessing information from Galaxy

a) How big is the largest RefSeq gene on human chromosome 21? Solve this problem by using Galaxy.

(1) First go to Galaxy (http://usegalaxy.org). Optionally, you can register (under the "User" tab).

(2) On the left sidebar, choose "Get Data" then "UCSC Main table browser."

(3) Set the clade (Mammal), genome (Human), assembly (GRCh37 [or try GRCh38]), group (Genes and Gene Prediction Tracks), track (RefSeq Genes), table (RefGene), region (click position then enter "chr21" without the quotation marks) then click "lookup" right next to the position. Under output format choose "BED-browser extensible data" and click the box "Send output to Galaxy."

(4) Optionally, click "summary/statistics" to get a quick look at how many proteins are assigned to chromosome 21. (That answer is currently 636.)

(5) At the lower left part of the page, click "get output." Note that you now have a variety of output options; choose BED and click "Send query to Galaxy."

(6) Galaxy's central panel informs you that the job is added to the queue.

(7) Your data set is available in the history panel to the right. Click the data set header [1: UCSC Main on Human: refGene (chr21:1-46944323)] to see the number of regions and to see the column headers. Click the "eye" icon to see your data in the central panel.

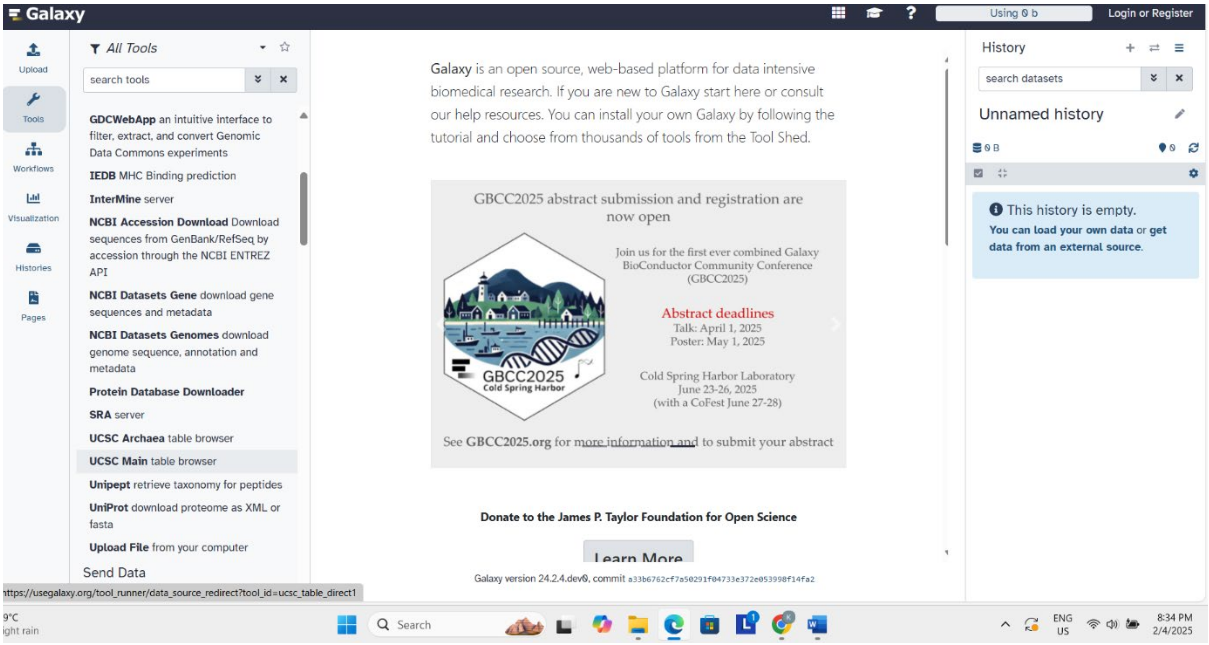
(8) Next figure out the size of the genes. First, add a new column. On the left Galaxy panel click "Text Manipulation" then "Compute an expression on every row." Add the expression c3-c2 to take the end position of each gene and subtract the beginning. For "Round result?" choose "Yes." Click "Execute."

(9) A new data set is created, called "Compute on data 1." There is a new column 13 with the sizes of all the genes. Go to the left sidebar of Galaxy, click "Filter and Sort," click "Sort data in ascending or descending order" and choose the query; the column (c13); the flavor (numerical sort); the order (descending); and click Execute.

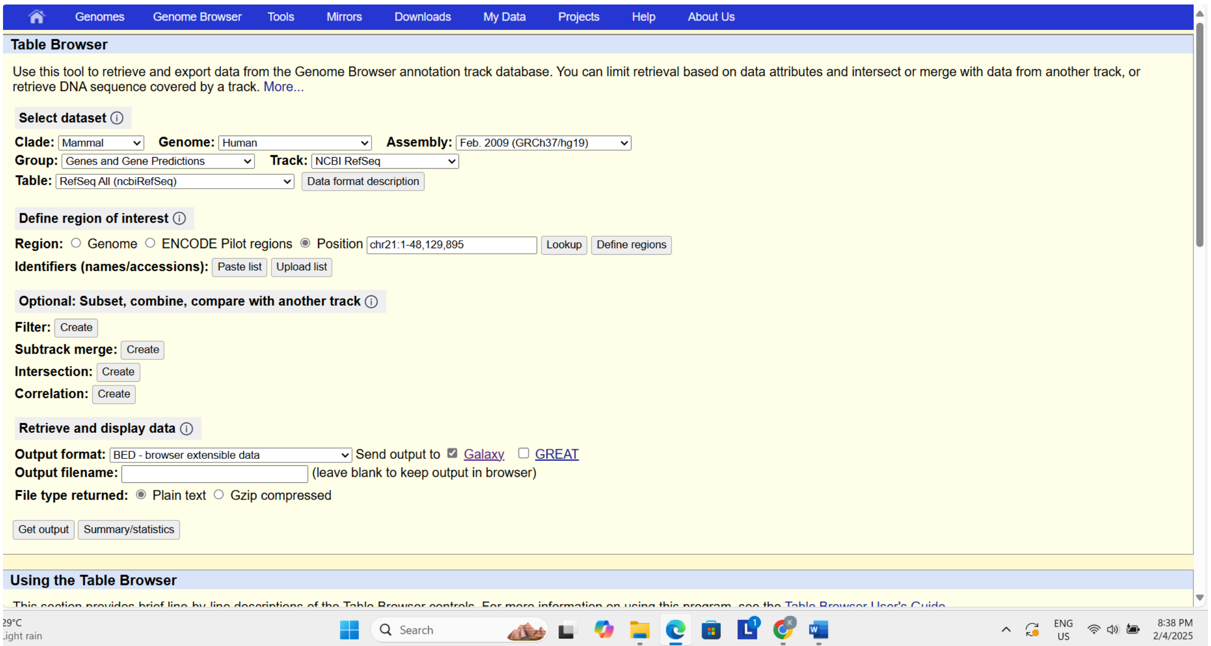
(10) A new data set is created. Click the eye icon to see your spreadsheet in the main Galaxy panel. Your answer is there on the first (top) row. Optionally, go to "Text Manipulation," select "Cut columns from a table," and Cut columns (c5,c6,c7,c8,c9,c10,c11,c12). This will clean up your table, making it easier to see column 13 with the gene lengths.

10.Answer:

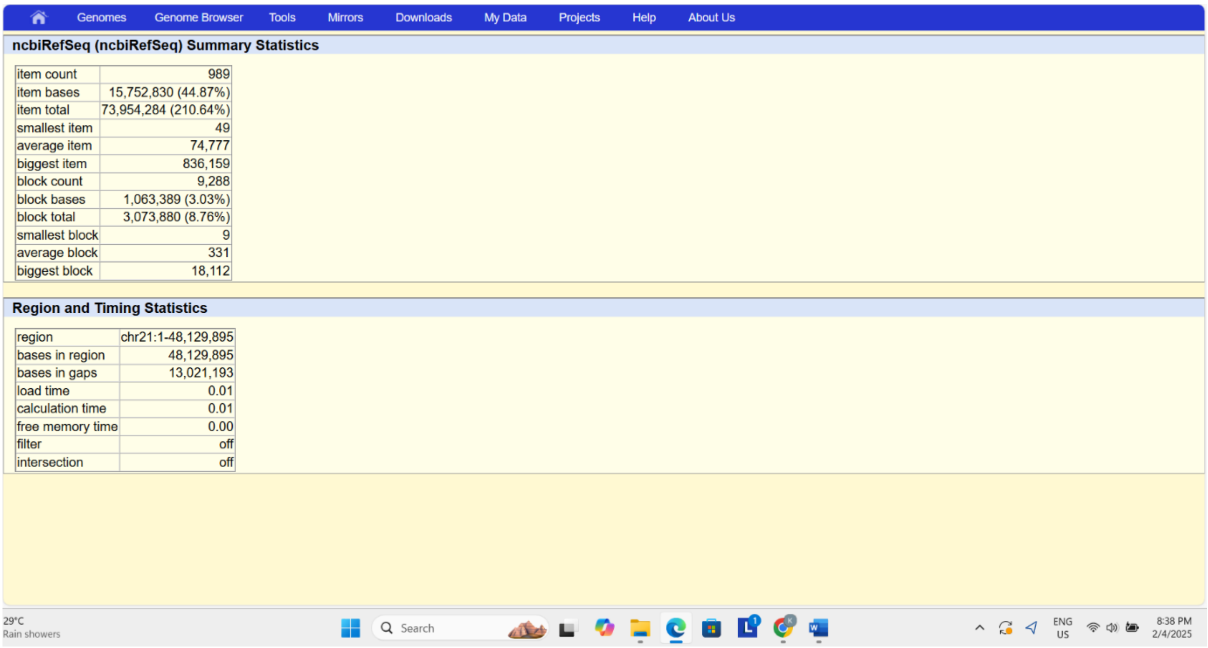
Here is the Galaxy home page:



Set the Table Browser entries as follows:



The current total number of genes (989) on this chromosome is shown by clicking the summary/statistics tab:



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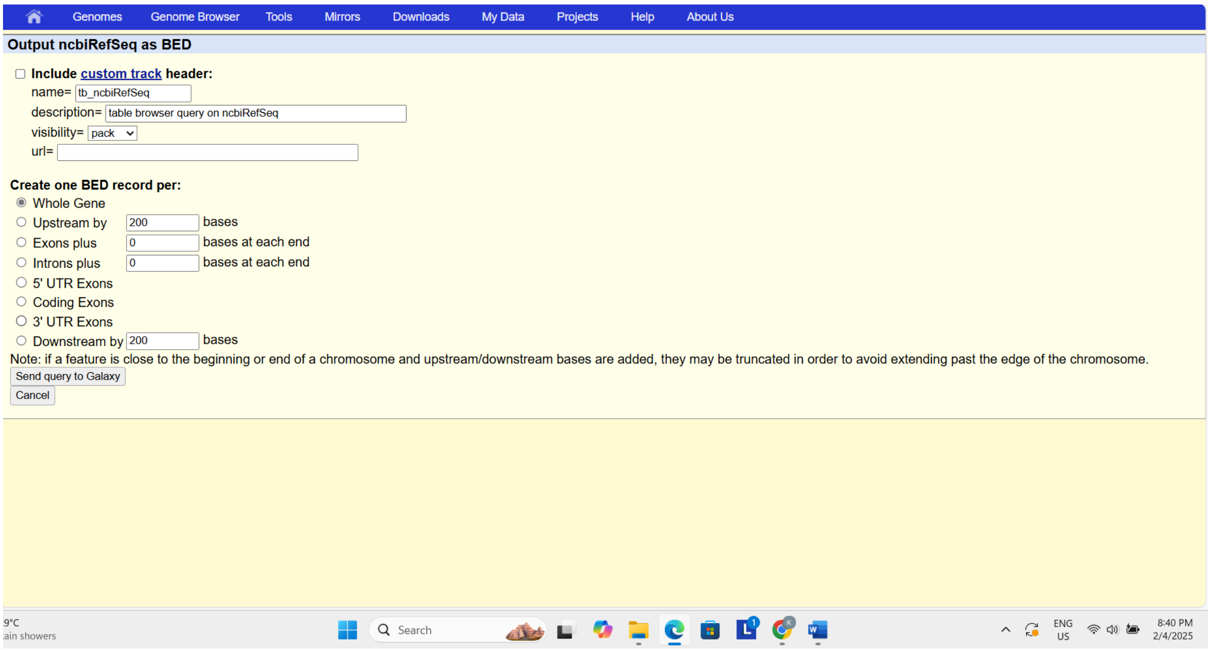
ncbiRefSeq (ncbiRefSeq) Summary Statistics

item count	989
item bases	15,752,830 (44.87%)
item total	73,954,284 (210.64%)
smallest item	49
average item	74,777
biggest item	836,159
block count	9,288
block bases	1,063,389 (3.03%)
block total	3,073,880 (8.76%)
smallest block	9
average block	331
biggest block	18,112

Region and Timing Statistics

region	chr21:1-48,129,895
bases in region	48,129,895
bases in gaps	13,021,193
load time	0.01
calculation time	0.01
free memory time	0.00
filter	off
intersection	off

Click “get ouput” then send query to Galaxy.



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Output ncbiRefSeq as BED

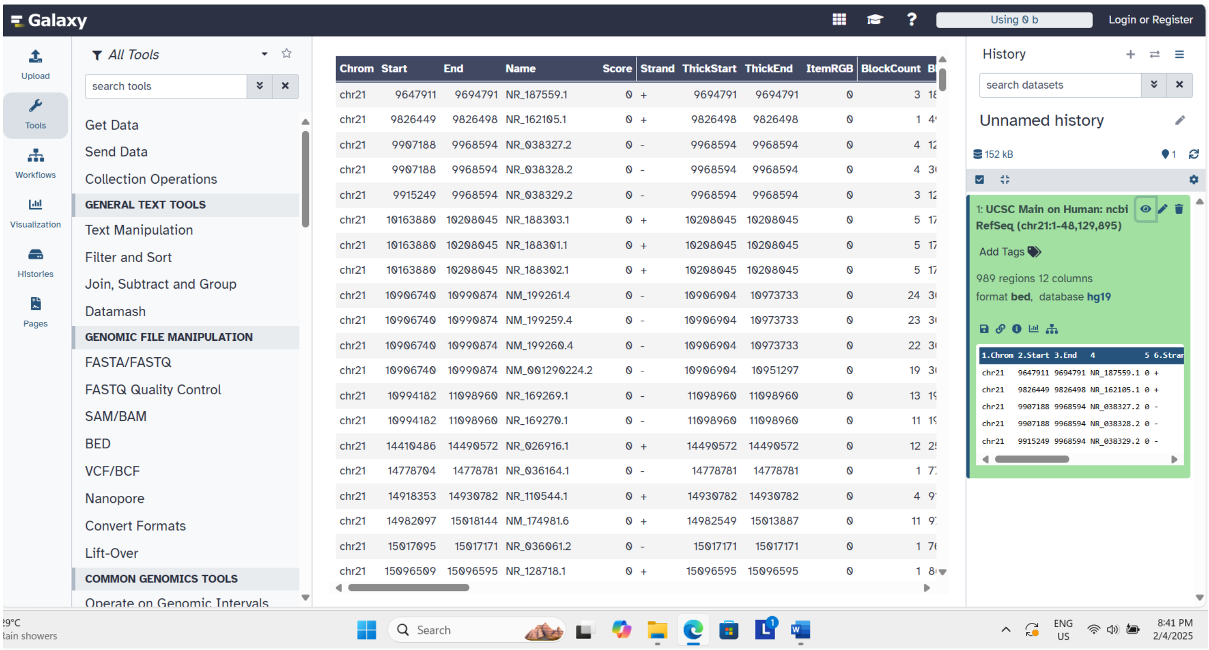
☐ **Include custom track header:**
name= tb_ncbiRefSeq
description= table browser query on ncbiRefSeq
visibility= pack
url=

Create one BED record per:
- ◉ Whole Gene
- ○ Upstream by 200 bases
- ○ Exons plus 0 bases at each end
- ○ Introns plus 0 bases at each end
- ○ 5' UTR Exons
- ○ Coding Exons
- ○ 3' UTR Exons
- ○ Downstream by 200 bases

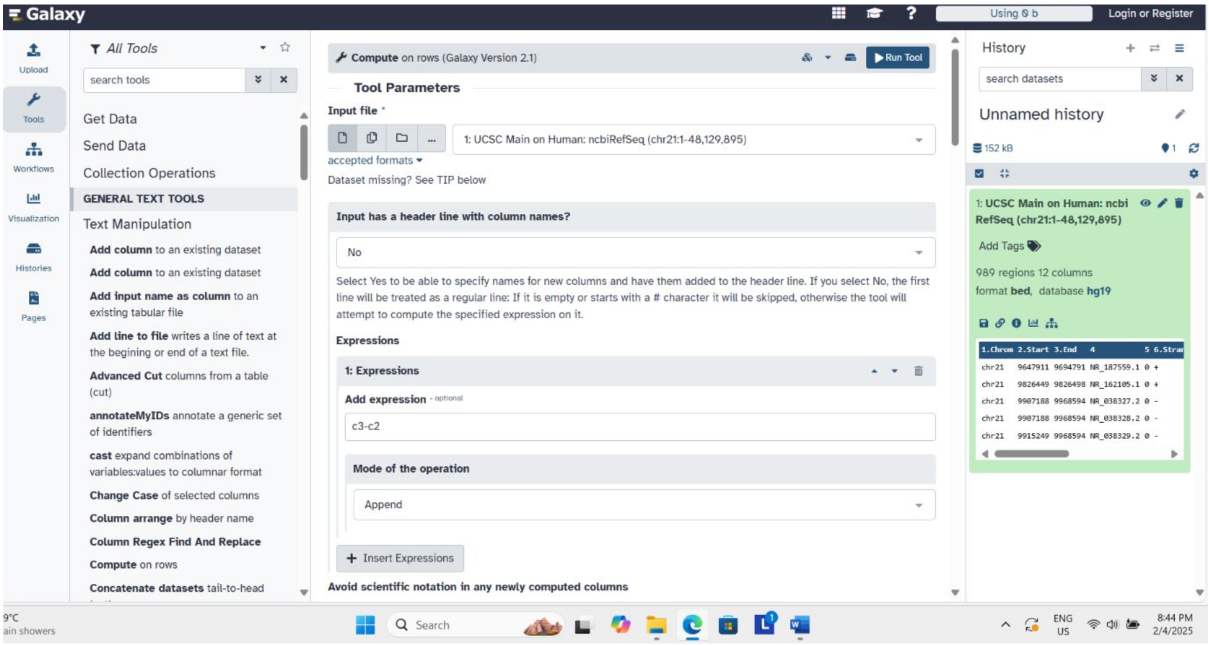
Note: if a feature is close to the beginning or end of a chromosome and upstream/downstream bases are added, they may be truncated in order to avoid extending past the edge of the chromosome.

Send query to Galaxy
Cancel

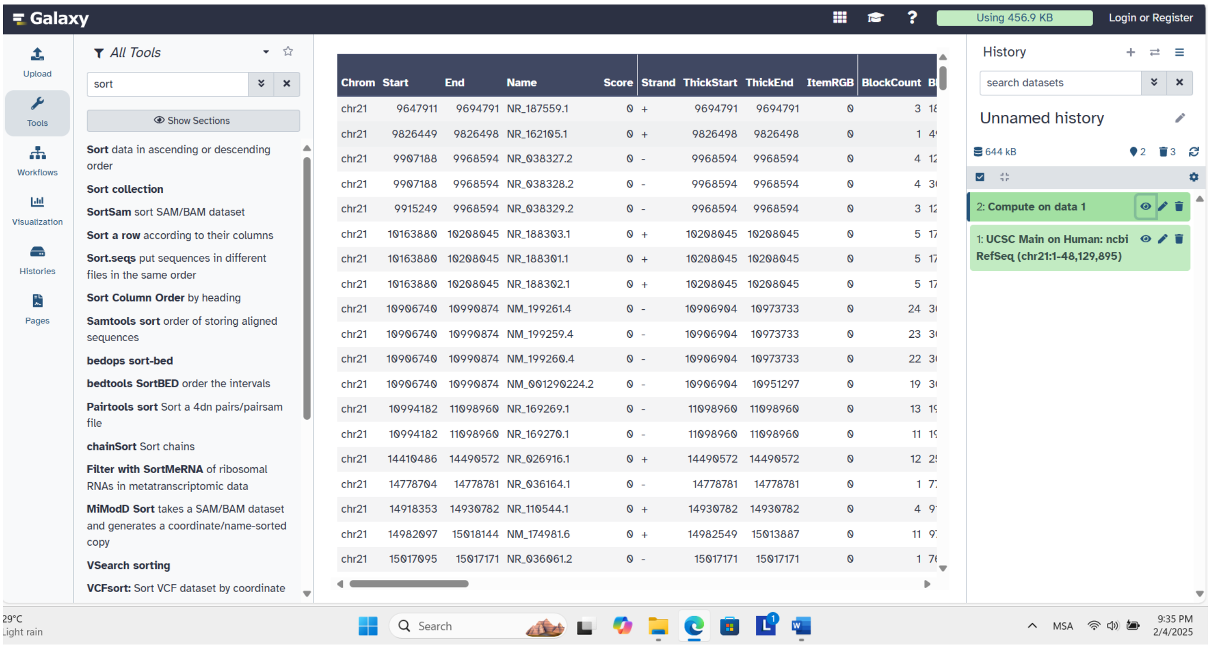
Follow the above directions. Imported data appear in the right sidebar; click the eye icon to see the data in the main display:



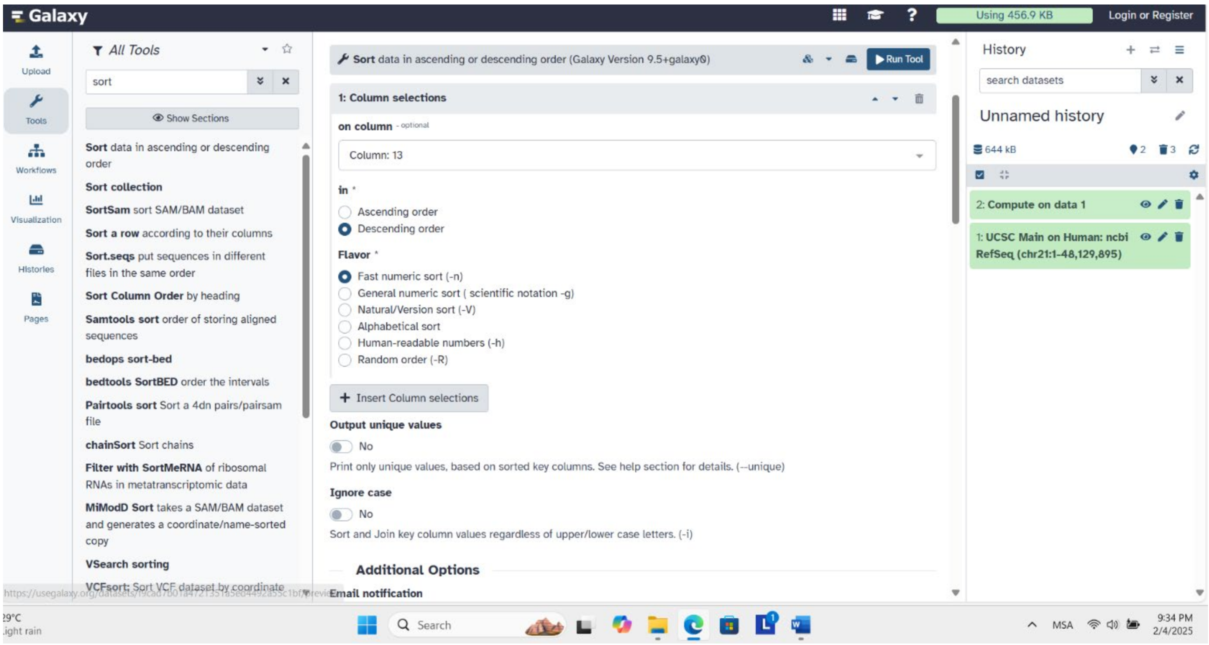
As indicated in the problem's directions (above), use Tools > Text Manipulation > Compute on row > add expression c3-c2 (this will take the end position minus the start position of every gene on chromosome 21).



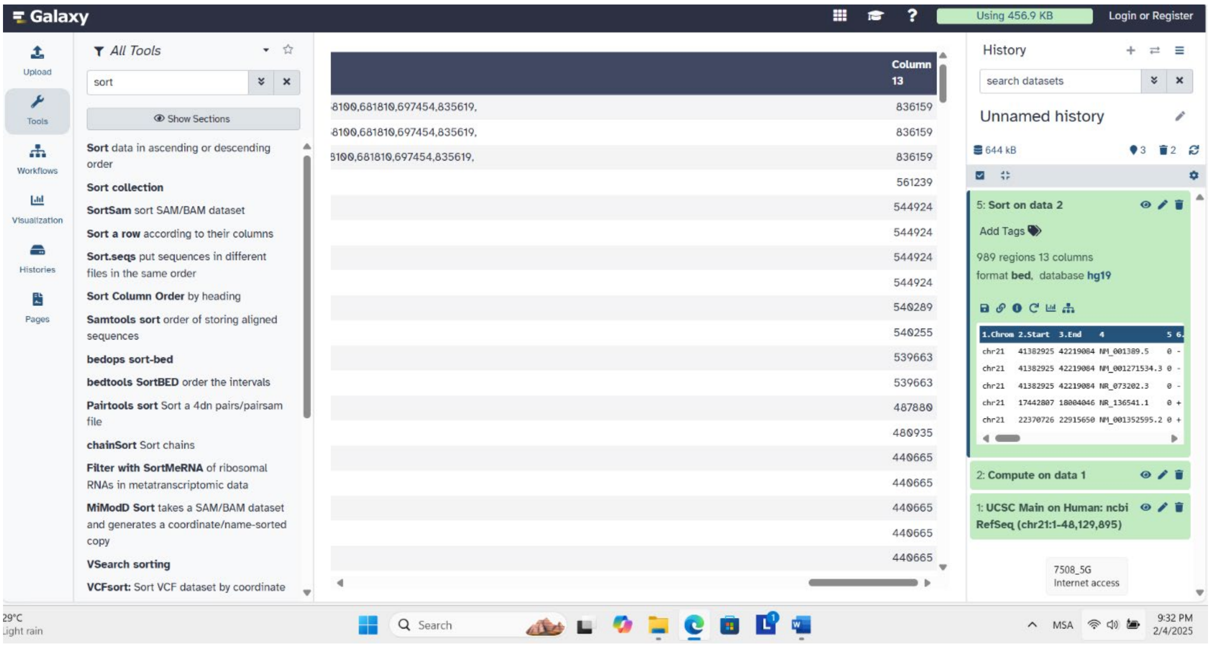
Here is the output: a new file is created (data set 4: compute on data 3) with a new column 13 showing the size of every gene:



Now we need to sort the results on column 13 to see which is the largest gene on chromosome 21. Tools > Filter and sort. Execute.



The answer is that the largest gene is 836159 base pairs.



Galaxy
Using 456.9 KB
Login or Register
Upload
Tools
Workflows
Visualization
Histories
Pages
All Tools
sort
Show Sections
Sort data in ascending or descending order
Sort collection
SortSam sort SAM/BAM dataset
Sort a row according to their columns
Sort.seqs put sequences in different files in the same order
Sort Column Order by heading
Samtools sort order of storing aligned sequences
bedops sort-bed
bedtools SortBED order the intervals
Pairtools sort Sort a 4dn pairs/pairsam file
chainSort Sort chains
Filter with SortMeRNA of ribosomal RNAs in metatranscriptomic data
MiModD Sort takes a SAM/BAM dataset and generates a coordinate/name-sorted copy
VSearch sorting
VCFsort: Sort VCF dataset by coordinate
Column 13
History
search datasets
Unnamed history
644 kB
5: Sort on data 2
Add Tags
989 regions 13 columns
format bed, database hg19
2: Compute on data 1
1: UCSC Main on Human: ncbi RefSeq (chr21:1-48,129,895)